

Logic

Aria Ruler(Forouzandeh)

May 13, 2026

This page intentionally left blank.

Contents

1	Atomes	1
1.1	Basic tools	1
2	propositional logic (zeroth-order logic)	5
2.1	What is proposition?	5
2.2	Rules of propositions.	6
3	first-order logic	13
3.1	what is formula?	13
3.2	Rules of first order logic.	16
4	The beginning of set theory	21
4.1	Axioms of set theory	21
5	Relations and Functions	31
5.1	Orderd pair and Cartesian product	31
5.2	Relations	33
5.3	Functions	36
5.4	Family of sets	40
5.5	Structures and morphisms	43
5.6	Equivalences and Partitions	44
5.7	Orderings	48
6	The beginning of Abstract Algebra	55
6.1	Group theory	55
6.2	Ring theory	55
7	Natural numbers	61
7.1	Introduction to Natural numbers	61
7.2	Properties of Natural numbers	62
7.3	The Recurtion theorem	63
7.4	Arithmetic of Natural numbers	66
7.5	Peano arithmetic	67
8	From Natural numbers to Complex numbers	69
8.1	Integers“ $\mathbb{Z}$ ”	69

8.2	Rational numbers “ $\mathbb{Q}$ ”	70
8.3	Real numbers “ $\mathbb{R}$ ”	72
8.4	Complex numbers “ $\mathbb{C}$ ”	74
9	Finite and Infinite sets	77
9.1	Equipotent and Order in sets	77
9.2	Finite sets	78
9.3	Countable sets	78
9.4	Uncountable sets	79
10	Ordinals	81
10.1	Well ordered sets	81
10.2	Ordinal numbers	82
10.3	Transfinite Induction	82
10.4	Transfinite Recursion	82
10.5	Arithmetic of Ordinals	83
10.6	Hyper Peano arithmetic	83
11	Cardinals	87
12	Axiom of choice	89
13	hyperreal numbers(${}^*\mathbb{N}, {}^*\mathbb{Z}, {}^*\mathbb{Q}, {}^*\mathbb{R}$) and surreal numbers	91
14	fantasy numbers($\mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord), \mathbb{C}(Ord)$)	95
14.1	Hyper Integers “ $\mathbb{Z}(Ord)$ ”	95
14.2	Hyper Rational numbers “ $\mathbb{Q}(Ord)$ ”	96
14.3	Hyper Real numbers “ $\mathbb{R}(Ord)$ ”	98
14.4	Hyper Complex numbers “ $\mathbb{C}(Ord)$ ”	100
15	New beginning for calculus	103
15.1	Derivatives	103
16	Lebesgue Integral	105
17	New beginning for Probability theory	107

Chapter 1

Atomes

1.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 1.1.1. *Urelements* are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.

In this book, we will denote individual urelements using lowercase letters $u, v, w, \dots$

Undefined 1.1.2. *Type* parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).

1.	u			
2.		v		
3.			w	

Rule 1.1.1. *fusion* it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)

$m.$	u_0	
	$\vdots$	
$m + n.$	u_n	
$o.$		$\{u_0, u_1 \dots u_n\}$

Undefined 1.1.3. Set we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).

definition 1.1.1. Object urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).

Rule 1.1.2. fission it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)

$o.$		$\{a_0, a_1 \dots a_n\}$
$m.$	a_0	
	$\vdots$	
$m + n.$	a_n	

Example 1.1.1. $\{a, b, c\} / \therefore \{a, b\}$ (prove if $\{a, b, c\}$ exists then $\{a, c\}$ exists)

1.		$\{a, b, c\}$
2.	a	$fi1$
3.	b	$fi1$
4.	c	$fi1$
5.		$\{a, c\} \quad fu2,3,4$

Example 1.1.2. $\{a, b\} / \therefore \{b, a\}$

1.		$\{a, b\}$
2.	a	$fi1$
3.	b	$fi1$
4.		$\{b, a\} \quad fu2,3$

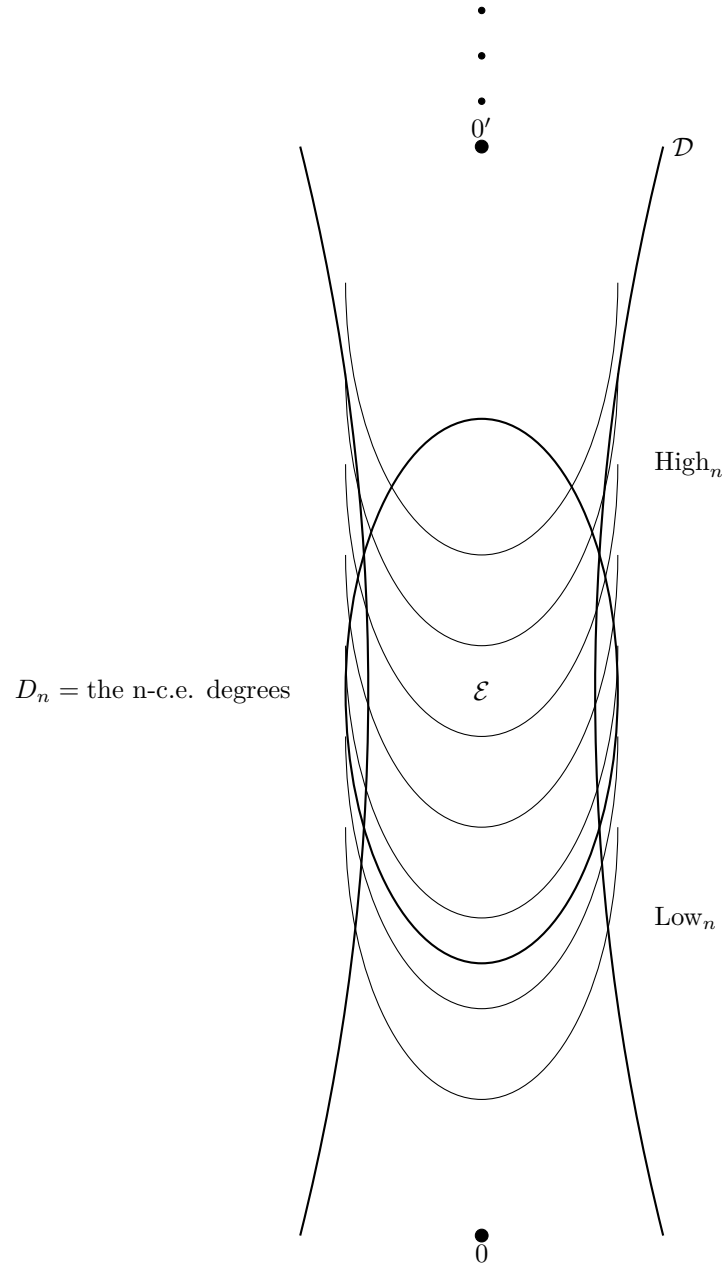


Figure 1.1: Schematic of the n-c.e. hierarchy and degree structure.

Chapter 2

propositional logic (zeroth-order logic)

2.1 What is proposition?

definition 2.1.1. predicate “ $\text{set}()$ ” and identity “ $=$ ” and membership “ $\in$ ” symbols are predicates. “ $\text{set}()$ ” is unary and identity and membership are binary.

capital letter “ (P, Q, R) ” shows **predicates**.

definition 2.1.2. Atomic proposition we define atomic propositions like this.

1. if “ a ” is an obj then “ $\text{set}(a)$ ” is an atomic proposition.
2. if “ a, b ” are obj then “ $a = b$ ” is an atomic proposition.
3. if “ a, b ” are obj then “ $a \in b$ ” is an atomic proposition.

we denote atomic propositions with “ p_i ”.

Undefined 2.1.1. logical operators ($\neg, \rightarrow, \wedge, \vee$) tool that used between atoms to make more complex propositions (all of the opratores are binary except “ $\neg$ ” that is unary).

you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.

definition 2.1.3. Propositions these are propositions.

1. atomic propositions “ (p_i) ”.
2. if “ p ” is a formula then “ $\neg p$ ” is a formula too.
3. if “ p, q ” are two propositions then “ $p \bigcirc q$ ” is a formula too. ($\bigcirc$ can be “ $\rightarrow, \wedge, \vee$ ”).

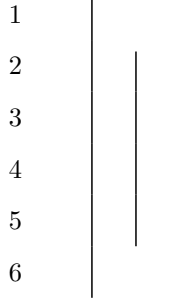
for example “ $(p_0 \rightarrow p_1, \neg p_3)$ ”

we denote propositions with “ $p, q, r, \dots$ ”.

2.2 Rules of propositions.

definition 2.2.1. *proposition schema(variable proposition)* a proposition that is unknown for us and it works as variable.(we show them with “ $\varphi, \psi, \theta, \dots$ ”)

Undefined 2.2.1. *Assumption for conditional proof(acp)* is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shorter than left one.(it can't fall out). and every hypothesis that opens must be closed.)



Rule 2.2.1. *rules of logical operations* ($\rightarrow I, \rightarrow E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \perp I, \perp E$) this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)

φ $\vdots$ ψ $\therefore \varphi \rightarrow \psi \quad \rightarrow I$	$\varphi \rightarrow \psi$ $\frac{\varphi}{\therefore \psi} \quad \rightarrow E$
---	--

Example 2.2.1. for “ p, q, r ” we can prove these.

1. $p \rightarrow q, q \rightarrow r / \therefore p \rightarrow r$

2. $q / \therefore p \rightarrow q$

1	$p \rightarrow q$	ass
2	$q \rightarrow r$	ass
3	p	acp
4	q	$\rightarrow E1, 3$
5	r	$\rightarrow E2, 4$
6	$p \rightarrow r$	$\rightarrow I$

1	q	ass
2	p	acp
3	q	ass
4	$p \rightarrow q$	$\rightarrow I$

$\frac{\varphi \quad \psi}{\therefore \varphi \wedge \psi} \quad \wedge I$	$\frac{\varphi \wedge \psi}{\therefore \varphi} \quad \wedge E$	$\frac{\varphi \wedge \psi}{\therefore \psi} \quad \wedge E$
--	---	--

Example 2.2.2. for “ p, q, r ” we can prove these.

1. $p \wedge q / \therefore q \wedge p$

2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$

3. $p \rightarrow q / \therefore (p \wedge r) \rightarrow q$

4. $(p \wedge q) \rightarrow r / \therefore p \rightarrow (q \rightarrow r)$

5. $p \rightarrow (q \rightarrow r) / \therefore (p \wedge r) \rightarrow q$

1	$p \wedge q$	<i>ass</i>	1	$(p \wedge q) \wedge r$	<i>ass</i>
2	p	$\wedge E, 1$	2	$p \wedge q$	$\wedge E, 1$
3	q	$\wedge E, 1$	3	r	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$	4	p	$\wedge E, 2$
			5	q	$\wedge E, 2$
			6	$q \wedge r$	$\wedge I, 3, 5$
			7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \rightarrow q$	<i>ass</i>	1	$(p \wedge q) \rightarrow r$	<i>ass</i>	1	$p \rightarrow (q \rightarrow r)$	<i>ass</i>
2	$\left \begin{array}{l} p \wedge r \\ p \\ q \end{array} \right.$	<i>acp</i>	2	$\left \begin{array}{l} p \\ p \wedge q \\ r \end{array} \right.$	<i>acp</i>	2	$\left \begin{array}{l} p \wedge q \\ p \\ q \end{array} \right.$	<i>acp</i>
3		$\wedge E, 2$	3		$\wedge I, 2, 3$	3		$\wedge E, 2$
4		$\rightarrow E, 1, 3$	4		$\rightarrow E, 1, 4$	4		$\rightarrow E, 1, 3$
5	$(p \wedge r) \rightarrow q$	$\rightarrow I$	5		$\rightarrow I$	5		$\wedge E, 2$
			6	$q \rightarrow r$	$\rightarrow I$	6	r	$\rightarrow E, 4, 5$
			7	$p \rightarrow (q \rightarrow r)$	$\rightarrow I$	7	$(p \wedge q) \rightarrow r$	$\rightarrow I$

$\frac{\varphi}{\therefore \varphi \vee \psi} \quad \vee I \qquad \frac{\varphi}{\therefore \psi \vee \varphi} \quad \vee I$	$\begin{array}{c} \varphi \vee \psi \\ \hline \varphi \\ \vdots \\ \theta \\ \hline \psi \\ \vdots \\ \theta \\ \therefore \theta \end{array} \quad \vee E$
--	---

Example 2.2.3. for “ p, q, r ” we can prove these.

1. $p \vee q / \therefore q \vee p$
2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$
3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$
4. $(p \rightarrow q) \wedge (r \rightarrow s) / \therefore (p \vee r) \rightarrow (q \vee s)$

1	$p \vee q$	<i>ass</i>
2	$\mid p$	<i>acp</i>
3	$\mid p \vee q$	$\vee I, 2$
4	$\mid q$	<i>acp</i>
5	$\mid q \vee p$	$\vee I, 4$
6	$q \vee p$	$\vee E, 1$

1	$(p \vee q) \vee r$	<i>ass</i>
2	$\mid p \vee q$	<i>acp</i>
3	$\mid \mid p$	<i>acp</i>
4	$\mid \mid p \vee (q \vee r)$	$\vee I, 3$
5	$\mid \mid q$	<i>acp</i>
6	$\mid \mid q \vee r$	$\vee I, 5$
7	$\mid \mid p \vee (q \vee r)$	$\vee I, 6$
8	$\mid p \vee (q \vee r)$	$\vee E, 2$
9	$\mid r$	<i>acp</i>
10	$\mid q \vee r$	$\vee I, 9$
11	$\mid p \vee (q \vee r)$	$\vee I, 10$
12	$p \vee (q \vee r)$	$\vee E, 1$

1	$p \wedge (q \vee r)$	<i>ass</i>
2	p	$\wedge E1$
3	$q \vee r$	$\wedge E1$
4	q	<i>acp</i>
5	$p \wedge q$	$\wedge E2, 4$
6	$(p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	r	<i>acp</i>
8	$p \wedge r$	$I\wedge 2, 7$
9	$(p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$(p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \rightarrow q) \wedge (r \rightarrow s)$	<i>ass</i>
2	$(p \rightarrow q)$	$\wedge E1$
3	$(r \rightarrow s)$	$\wedge E1$
4	$p \vee r$	<i>acp</i>
5	p	<i>acp</i>
6	q	$\rightarrow E2, 5$
7	$q \vee s$	$\vee I, 6$
8	r	<i>acp</i>
9	s	$\rightarrow E3, 8$
10	$q \vee s$	$\vee I9$
11	$q \vee s$	$\vee E, 4$
12	$(p \vee r) \rightarrow (q \vee s)$	$\rightarrow I$

definition 2.2.2. $\perp : \varphi \wedge \neg\varphi$

$\begin{array}{c} \varphi \\ \vdots \\ \perp \end{array}$ $\therefore \neg\varphi \quad \neg I$	$\begin{array}{c} \neg\varphi \\ \vdots \\ \perp \end{array}$ $\therefore \varphi \quad \neg E$
---	---

definition 2.2.3. schema of prove you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form. you can't prove it directly. here we use a technique calls "**schema of prove**". it let us prove for all situations at the same time.

Theorem 2.2.1. (schema) ($\perp I, \perp E$)

$\begin{array}{c} \varphi \\ \neg\varphi \\ \hline \therefore \perp \end{array} \quad \perp I$	$\frac{\perp}{\therefore \varphi} \quad \perp E$
--	--

1	φ	<i>ass</i>
2	$\neg\varphi$	<i>ass</i>
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\perp$	<i>def.3</i>

1	$\perp$	<i>ass</i>
2	$\neg\varphi$	<i>acp</i>
3	$\vdots$	
4	$\perp$	<i>rp1</i>
5	φ	$\neg E$

Attention 2.2.1. *Intuitionists can skip the “ $\neg E$ ” rule that is the “RAA” rule in old notation and use “ $\perp E$ ” as a rule instead.*

Example 2.2.4. *for “ p, q, r ” we can prove these.*

$$1. p \rightarrow q, \neg q / \therefore \neg p$$

$$2. p \rightarrow q / \therefore \neg p \vee q$$

1	$p \rightarrow q$	ass
2	$\neg q$	ass
3	p	acp
4	q	$\rightarrow E, 1, 3$
5	$\perp$	$\perp I, 2, 4$
6	$\neg p$	$\neg I$

1	$p \rightarrow q$	ass
2	$\neg(\neg p \vee q)$	acp
3	p	acp
4	q	$\rightarrow E, 1, 3$
5	$\neg p \vee q$	$\vee I, 4$
6	$\perp$	$\perp I, 2, 5$
7	$\neg p$	$\neg I$
8	$\neg p \vee q$	$\vee I, 7$
9	$\perp$	$\perp I, 2, 8$
10	$\neg p \vee q$	$\neg I$

definition 2.2.4. $\varphi \leftrightarrow \psi : (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$

Theorem 2.2.2. (*schema*) $(\leftrightarrow I, \leftrightarrow E)$

φ $\vdots$ ψ ψ $\vdots$ φ $\therefore \varphi \leftrightarrow \psi \quad \leftrightarrow I$	<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="text-align: center; padding: 5px;"> $\varphi \leftrightarrow \psi$ $\frac{\varphi}{\therefore \psi}$ </td><td style="text-align: center; padding: 5px;"> $\varphi \leftrightarrow \psi$ $\frac{\psi}{\therefore \varphi}$ </td></tr> <tr> <td style="text-align: center; padding: 5px;">$\leftrightarrow E$</td><td style="text-align: center; padding: 5px;">$\leftrightarrow E$</td></tr> </table>	$\varphi \leftrightarrow \psi$ $\frac{\varphi}{\therefore \psi}$	$\varphi \leftrightarrow \psi$ $\frac{\psi}{\therefore \varphi}$	$\leftrightarrow E$	$\leftrightarrow E$
$\varphi \leftrightarrow \psi$ $\frac{\varphi}{\therefore \psi}$	$\varphi \leftrightarrow \psi$ $\frac{\psi}{\therefore \varphi}$				
$\leftrightarrow E$	$\leftrightarrow E$				

1	φ	<i>acp</i>		
	$\vdots$			
n	ψ			
$n+1$	$\varphi \rightarrow \psi$	$\rightarrow, I$	1	$\varphi \leftrightarrow \psi$ <i>ass</i>
$n+2$	ψ	<i>acp</i>	2	φ <i>ass</i>
	$\vdots$		3	$(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$ <i>def.1</i>
m	φ		4	$\varphi \rightarrow \psi$ $\wedge E3$
$m+1$	$\psi \rightarrow \varphi$	$\rightarrow, I$	5	ψ $\rightarrow E1, 4$
$m+2$	$(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$	$\wedge I$		
$m+3$	$\varphi \leftrightarrow \psi$	<i>def.</i> , $m+3$		

Exercise 2.2.1. for “ p, q, r ” prove.

1. $p / \therefore p$
2. $/ \therefore p \vee \neg p$
3. $\neg p, p \vee q / \therefore q$
4. $\neg p / \therefore p \rightarrow q$
5. $p \rightarrow q \therefore / \therefore \neg q \rightarrow \neg p$
6. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$
7. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$ (*De morgan*)
8. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$ (*De morgan*)
9. $(\neg p \rightarrow q) / \therefore ((p \rightarrow q) \rightarrow q)$
10. $/ \therefore (p \wedge q \rightarrow r) \wedge (p \wedge s \rightarrow r) \rightarrow (p \wedge (q \vee s) \rightarrow r)$
11. $/ \therefore (p \wedge q \rightarrow r) \vee (p \wedge s \rightarrow r) \rightarrow (p \wedge (q \wedge s) \rightarrow r)$
12. $/ \therefore (p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
13. $/ \therefore (p \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (p \rightarrow r))$

Attention 2.2.2. we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves. so they are all rules.

This page intentionally left blank.

Chapter 3

first-order logic

3.1 what is formula?

obj are two types. we know them and we know what exactly are we talking about and unknowns are mystery. all the obj we have used until now were known ones we call them "**constants**".

Undefined 3.1.1. variables *A variable is a symbol (usually a letter like x, y , or z) that represents an unknown obj.*

definition 3.1.1. Atomic formula *we define atomic formulas like this.*

1. if " x " is an obj (constant or variable) then " $\text{Set}(x)$ " is an atomic formula.
2. if " x, y " are obj (constant or variable) then " $x = y$ " is an atomic formula.
3. if " x, y " are obj (constant or variable) then " $x \in y$ " is an atomic formula.

we denote atomic formulas with " A_i ".

definition 3.1.2. well formed formula (WWF) *these are formulas.*

1. atomic formulas (A_i).
2. if A is a formula then $\neg A$ is a formula too.
3. if A, B are two formulas then $A \circ B$ is a formula too. ($\circ$ can be $\rightarrow, \wedge, \vee$).

for example ($A_0 \rightarrow A_1, \neg A_3$)

we easily call WWF "formula" and also we denote WWF with $A, B, C, \dots$

definition 3.1.3. open formula *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call "**closed formulas**" or "**sentence**". Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.

definition 3.1.4. Substitution for obj *we define Substitution an obj with variable like this.*

1. for variables if $x = y$ then $y[a/x] := a$ and if $x \neq y$ then $y[a/x] := y$.

2. for constants $c[a/x] := c$.

definition 3.1.5. Substitution for formulas we define Substitution for a formula like this.

1. for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like $x = y$ or $x \in y$ (x, y can be both constants and variables). we define it like this.

$$\text{Set}(y)[a/x] := \text{Set}(y[a/x])$$

$$(x = y)[a/z] := (x[a/z] = y[a/z])$$

$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. if formula A is complex we define it like this.

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ can be $\rightarrow, \wedge, \vee$).

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x . we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 3.1.6. Open proposition assume formula " A " with variables " $x_0, x_1, \dots, x_n$ " in it and a set " U ". we say formula " A " is a "open proposition in U " when for every obj " $a_0, a_1, \dots, a_n$ " in " U "

$$A([a_0/x_0][a_1/x_1] \dots [a_n/x_n])$$

is a proposition we call it open proposition and denote it with " $(P(x), Q(x, y), \dots)$ "

definition 3.1.7. Big "AND" and "OR" assume a set like " $\{a_0, a_1, a_2, \dots, a_n\}$ " we define big "AND" like this.

$$\bigwedge_{i=0}^n P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge \dots \wedge P(a_n)$$

and big "OR"

$$\bigvee_{i=0}^n P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee \dots \vee P(a_n)$$

definition 3.1.8. Universe of discourse we specify a set like " $\{a_0, a_1, a_2, \dots, a_n\}$ " as universe of a Logic world.

Rule 3.1.1. use of open propositions assume universe of discourse is " $\{a_0, a_1, a_2, \dots, a_n\}$ "

$$c \quad P(x)$$

means :

$$\begin{array}{ll}
c_0 & P[a_0/x] \\
c_1 & P[a_1/x] \\
c_2 & P[a_2/x] \\
& \vdots \\
c_n & P[a_n/x]
\end{array}$$

if Px was been assemed then like this

$$c \quad \left| \begin{array}{l} P(x) \\ \vdots \end{array} \right.$$

means :

$$\begin{array}{ll}
c_0 & \left| \begin{array}{l} P[a_0/x] \\ \vdots \end{array} \right. \\
c_1 & \left| \begin{array}{l} P[a_1/x] \\ \vdots \end{array} \right. \\
c_2 & \left| \begin{array}{l} P[a_2/x] \\ \vdots \end{array} \right. \\
c_n & \left| \begin{array}{l} P[a_n/x] \\ \vdots \end{array} \right.
\end{array}$$

definition 3.1.9. predicate schema(variable predicate) a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, \dots$)

definition 3.1.10. Universal quantification assume universe of discourse is " $\{a_0, a_1, a_2, \dots, a_n\}$ " we define " $\forall x P(x)$ "

$$\forall x \phi(x) : \bigwedge_{i=0}^n \phi[a_i/x]$$

definition 3.1.11. Existential quantification assume universe of discourse is " $\{a_0, a_1, a_2, \dots, a_n\}$ " we define " $\exists x P(x)$ "

$$\exists x \phi(x) : \bigvee_{i=0}^n \phi[a_i/x]$$

as you can see " $\forall x \phi(x)$ " and " $\exists x \phi(x)$ " are sentences.

definition 3.1.12. *Free variables* variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sencece is not necessary a proposition)

3.2 Rules of first order logic.

Theorem 3.2.1. (schema) $(\forall I, \forall E, \exists I, \exists E)$ assume universe of discourse is “ $\{a_0, a_1, a_2, \dots, a_n\}$ ”

Rules	Terms and Conditions
$\frac{\phi(x)}{\therefore \forall x \phi(x)} \quad \forall I$	1. x is a variable. 2. $\phi(x)$ must not be assumed.
$\frac{\forall x \phi(x)}{\therefore \phi[a/x]} \quad \forall E$	1. a is an obj in universe of discourse.(it can be constant or variable.)
$\frac{\phi[a/x]}{\therefore \exists x \phi(x)} \quad \exists I$	1. a is an obj in universe of discourse.(it can be conatant or variable.)
$\begin{array}{l} \exists x \phi(x) \\ \left \begin{array}{l} \phi(x) \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \exists E$	1. x is a variable. 2. $\phi(x)$ must not be assumed in previous lines.(except ones that are closed) 3. x is not free in θ

1	$\phi(x)$	<i>ass</i>
2	$\phi[a_0/x]$	<i>def.1</i>
	$\vdots$	
n	$\phi[a_n/x]$	<i>def.1</i>
$n+1$	$\bigwedge_{i=0}^n \phi[a_i/x]$	$\wedge I, 2, 3, \dots, n+2$
$n+2$	$\forall x \phi(x)$	<i>def.n+3</i>

1	$\forall x \phi(x)$	<i>ass</i>
2	$\bigwedge_{i=0}^n \phi[a_i/x]$	<i>def.1</i>
3	$\phi[a/x]$	$\wedge E2$

1	$\phi[a/x]$	
2	$\bigvee_{i=0}^n \phi[a_i/x]$	$\vee I1$
3	$\exists x \phi(x)$	<i>def2</i>

1	$\exists x \phi(x)$	
2	$\bigvee_{i=0}^n \phi[a_i/x]$	
3	$\phi[a_0/x]$	<i>acp</i>
	$\vdots$	
n	θ	
$n+1$	$\phi[a_1/x]$	<i>acp</i>
	$\vdots$	
m	θ	
	$\vdots$	
l	$\phi[a_2/x]$	<i>acp</i>
	$\vdots$	
r	θ	
$r+1$	θ	$\exists E, 3$

Rule 3.2.1. (*schema*) ($= I, = E$)

$\therefore a = a$	$= I$	$a = b$	$a = b$
		$\frac{\phi(a)}{\therefore \phi(b)}$	$\frac{\phi(a)}{\therefore \phi(a)}$
		$= E$	$= E$

definition 3.2.1. *Uniqueness* we define " $\exists! x \phi(x)$ "

$$\exists! x \phi(x) : \exists x (\phi(x) \wedge \forall y (\phi(y) \rightarrow x = y))$$

Example 3.2.1. for P, Q, R we can prove these. Proof of last ones Is Left As An Exercise To The Reader. 😊

1. $\forall x (P(x) \wedge Q(x)) \therefore / \therefore \forall x P(x) \wedge \forall x Q(x)$ ($D F \forall \wedge$) (Distribution and Factorization of $\forall \wedge$)
2. $\exists x (P(x) \vee Q(x)) \therefore / \therefore \exists x P(x) \vee \exists x Q(x)$ ($D F \exists \vee$)

3. $\forall xP(x) \vee \forall xQ(x) / \therefore \forall x(P(x) \vee Q(x))$ ($F \forall \vee$)

4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists xP(x) \wedge \exists xQ(x)$ ($D \exists \wedge$)

5. $\forall x(P(x) \rightarrow Q(x)) / \therefore \forall xP(x) \rightarrow \forall xQ(x)$ ($D \forall \rightarrow$)

6. $\exists xP(x) \rightarrow \exists xQ(x) / \therefore \exists x(P(x) \rightarrow Q(x))$ ($F \exists \rightarrow$)

1	$\forall x(P(x) \wedge Q(x))$	<i>ass</i>	1	$\forall xP(x) \wedge \forall xQ(x)$	<i>ass</i>
2	$P(x) \wedge Q(x)$	<i>acp</i>	2	$\forall xP(x)$	$\wedge E$
3	$P(x)$	$\wedge E$	3	$\forall xQ(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$	4	$P(x)$	$\forall E$
5	$\forall xP(x)$	$\forall I$	5	$Q(x)$	$\forall E$
6	$\forall xQ(x)$	$\forall I$	6	$P(x) \wedge Q(x)$	$\wedge I$
7	$\forall xP(x) \wedge \forall xQ(x)$	$\wedge I$	7	$\forall x(P(x) \wedge Q(x))$	$\forall I$

1	$\exists x(P(x) \vee Q(x))$	<i>ass</i>	1	$\exists xP(x) \vee \exists xQ(x)$	<i>ass</i>
2	$P(x) \vee Q(x)$	<i>acp</i>	2	$\exists xP(x)$	<i>acp</i>
3	$P(x)$	<i>acp</i>	3	$P(x)$	<i>acp</i>
4	$\exists xP(x)$	$\exists I$	4	$P(x) \vee Q(x)$	$\vee I$
5	$\exists xP(x) \vee \exists xQ(x)$	$\vee I$	5	$\exists x(P(x) \vee Q(x))$	$\exists I$
6	$Q(x)$	<i>acp</i>	6	$\exists x(P(x) \vee Q(x))$	$\exists E$
7	$\exists xQ(x)$	$\exists I$	7	$\exists xQ(x)$	<i>acp</i>
8	$\exists xP(x) \vee \exists xQ(x)$	$\vee I$	8	$Q(x)$	<i>acp</i>
9	$\exists xP(x) \vee \exists xQ(x)$	$\vee E$	9	$P(x) \vee Q(x)$	$\vee I$
10	$\exists xP(x) \vee \exists xQ(x)$	$\exists E$	10	$\exists x(P(x) \vee Q(x))$	$\exists I$
			11	$\exists x(P(x) \vee Q(x))$	$\exists E$
			12	$\exists x(P(x) \vee Q(x))$	$\vee E$

1	$\forall xP(x) \vee \forall xQ(x)$	<i>ass</i>
2	$\forall xP(x)$	<i>acp</i>
3	$P(x)$	$\forall E$
4	$P(x) \vee Q(x)$	$\vee I$
5	$\forall x(P(x) \vee Q(x))$	$\forall I$
6	$\forall xQ(x)$	<i>acp</i>
7	$Q(x)$	$\forall E$
8	$P(x) \vee Q(x)$	$\vee I$
9	$\forall x(P(x) \vee Q(x))$	$\forall I$
10	$\forall x(P(x) \vee Q(x))$	$\vee E$

1	$\forall x(P(x) \rightarrow Q(x))$	<i>ass</i>
2	$\forall xP(x)$	<i>acp</i>
3	$P(x)$	$\forall E$
4	$P(x) \rightarrow Q(x)$	$\forall E$
5	$Q(x)$	$\rightarrow E$
6	$\forall xQ(x)$	$\forall I$
7	$\forall xP(x) \rightarrow \forall xQ(x)$	$\rightarrow I$

1	$\exists x(P(x) \wedge Q(x))$	<i>ass</i>
2	$P(x) \wedge Q(x)$	<i>acp</i>
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\exists xP(x)$	$\exists I$
6	$\exists xQ(x)$	$\exists I$
7	$\exists xP(x)$	$\exists E$
8	$\exists xQ(x)$	$\exists E$
9	$\exists xP(x) \wedge \exists xQ(x)$	$\wedge I$

1	$\exists xP(x) \rightarrow \exists xQ(x)$	$\rightarrow I$
2	$\exists xP(x) \vee \neg(\exists xP(x))$	<i>ex.</i>
3	$\exists xP(x)$	<i>acp</i>
4	$\exists Q(x)$	$\rightarrow I$
5	$Q(x)$	<i>acp</i>
6	$P(x) \rightarrow Q(x)$	<i>ex.</i>
7	$\exists x(P(x) \rightarrow Q(x))$	$\exists I$
8	$\exists x(P(x) \rightarrow Q(x))$	$\exists E$
9	$\neg(\exists xP(x))$	<i>acp</i>
10	$\forall x\neg P(x)$	<i>De morgan</i>
11	$\neg P(x)$	$\forall E$
12	$P(x) \rightarrow Q(x)$	<i>ex.</i>
13	$\exists x(P(x) \rightarrow Q(x))$	$\exists I$
14	$\exists x(P(x) \rightarrow Q(x))$	$\vee E$

Exercise 3.2.1. for P, Q, R prove.

1. $\forall xP(x) \wedge Q \therefore \therefore \forall x(P(x) \wedge Q)$
2. $\forall xP(x) \vee Q \therefore \therefore \forall x(P(x) \vee Q)$
3. $\exists xP(x) \wedge Q \therefore \therefore \exists x(P(x) \wedge Q)$
4. $\exists xP(x) \vee Q \therefore \therefore \exists x(P(x) \vee Q)$

5. $Q \rightarrow \forall xP(x) \therefore / \therefore \forall x(Q \rightarrow P(x))$
6. $Q \rightarrow \exists xP(x) \therefore / \therefore \exists x(Q \rightarrow P(x))$
7. $\forall xP(x) \rightarrow Q \therefore / \therefore \exists x(P(x) \rightarrow Q)$
8. $\exists xP(x) \rightarrow Q \therefore / \therefore \forall x(P(x) \rightarrow Q)$
9. $\exists x\neg P(x) \therefore / \therefore \neg\forall xP(x)$ (*De morgan*)
10. $\forall x\neg P(x) \therefore / \therefore \neg\exists xP(x)$ (*De morgan*)
11. $\exists x(P(x) \wedge \exists yQ(y) \rightarrow \forall zR(z)) / \therefore \forall y\forall z(\forall xP(x) \wedge Q(y) \rightarrow R(z))$
12. $\forall x\exists y(P(x) \wedge Q(y) \rightarrow \forall zR(z)) / \therefore \forall z(\exists xP(x) \wedge \forall yQ(y) \rightarrow R(z))$
13. $\forall xP(x) \rightarrow \exists yQ(y) / \therefore \exists x\exists y(P(x) \rightarrow Q(y))$
14. $\forall x\forall z\exists y\exists w(P(y, z) \vee Q(z) \rightarrow R(x, w)) / \therefore (\exists z\forall yP(y, z) \vee \exists zQ(z)) \rightarrow \forall x\exists wR(x, w)$
15. $\exists xP(x) \vee \exists x(Q(x) \wedge R(x)) / \therefore \exists x(P(x) \vee Q(x)) \wedge \exists x(P(x) \vee R(x))$
16. $\exists x\exists z\forall y(P(x, y) \rightarrow Q(z) \vee R(x, y)) / \therefore \forall x\exists yP(x, y) \rightarrow (\exists zQ(z) \vee \exists x\exists zR(x, y))$
17. $\forall x\exists y(\exists zP(x, y, z) \wedge Q(x, y)) \vee \forall x\exists y\exists z(P(x, y, z) \wedge R(x, y)) / \therefore \forall x\exists y\exists z(P(x, y, z) \wedge (Q(x, y) \vee R(x, y)))$

Attention 3.2.1. we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.

Chapter 4

The beginning of set theory

In language of set theory if we are talking about “ $\forall X(P(X))$ ” it means “ $\forall x(Set(x) \rightarrow P(x))$ ” and when we are talking about “ $\exists X(P(X))$ ” it means “ $\exists x(Set(x) \wedge P(x))$ ”. similarly if we are talking about “ $\forall x \in A \ P(x)$ ” it means “ $\forall x(x \in A \rightarrow P(x))$ ” and when we are talking about “ $\exists x \in A \ P(x)$ ” it means “ $\exists x(x \in A \wedge P(x))$ ”.

and we have a definitions.

$$Type_0(X) : \exists Y \quad X \in Y$$

$$Type_1(X) : \neg \exists Y \quad X \in Y$$

4.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 4.1.1. *Existence of universe of set theory* *There exists a set V_2 (we denote it as V_2 .it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition “ ϕ ” in “set theory” we mean proposition “ ϕ ” in “ V_2 ”).*

definition 4.1.1. *Subset* *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \rightarrow x \in B)$$

or i.e.

$$\forall x \in A \quad x \in B$$

Theorem 4.1.1. (schema) $(\subseteq I, \subseteq E)$

$\begin{array}{l} \left \begin{array}{l} x \in A \\ \vdots \\ x \in B \end{array} \right. \\ \therefore A \subseteq B \quad \subseteq I \end{array}$	$\begin{array}{l} A \subseteq B \\ \hline a \in A \\ \hline \therefore a \in B \quad \subseteq E \end{array}$
---	---

1	$x \in A$	<i>acp</i>	1	$A \subseteq B$	<i>ass</i>
	$\vdots$		2	$a \in A$	<i>ass</i>
n	$x \in B$		3	$\forall x(x \in A \rightarrow x \in A)$	<i>def.1</i>
$n+1$	$x \in A \rightarrow x \in B$	$I \rightarrow, 1, n$	4	$a \in A \rightarrow a \in B$	$\forall E, 4$
$n+2$	$\forall x(x \in A \rightarrow x \in B)$	$\forall I, n+1$	5	$a \in B$	$E \rightarrow 3, 5$
$n+3$	$A \subseteq B$	<i>def. n+2</i>			

Example 4.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	<i>ass</i>
2	$B \subseteq C$	<i>ass</i>
3	$x \in A$	<i>acp</i>
4	$x \in B$	$\subseteq E, 1, 3$
5	$x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 4.1.2. *Extensionality* Two sets are equal (are the same set) if they have the same elements.

$$\forall X, Y (\forall x (x \in X \leftrightarrow x \in Y) \rightarrow X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y ((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$$

Theorem 4.1.2. (*schema*) $(= I, = E)$

$\begin{array}{c l} x \in A \\ \vdots \\ x \in B \\ \hline x \in B \\ \vdots \\ x \in A \\ \hline \therefore A = B \end{array} = I$	$\begin{array}{cc} A = B & A = B \\ \frac{a \in A}{\therefore a \in B} & \frac{a \in B}{\therefore a \in A} \\ = E & = E \end{array}$
---	---

1	$x \in A$	acp
	$\vdots$	
n	$x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$x \in B$	acp
	$\vdots$	
m	$x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y ((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$	<i>Extensionality</i>
$m+3$	$(A \subseteq B \wedge B \subseteq A) \rightarrow A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \rightarrow, m+3, m+4$

Axiom 4.1.3. Weak comprehension (schema) this axioms says for every open proposition like “ $\phi(x, x_0, x_1, \dots, x_n)$ ” there is a set like “ X ” such that every “ x ” with “ $Type_0$ ” is a member of that set if and only if it holds “ $\phi(x, x_0, x_1, \dots, x_n)$ ”.

$$\forall x_0, x_1, \dots, x_n \exists X \forall x (Type_0(x) \wedge \phi(x, x_0, x_1, \dots, x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as “ $\{x \mid \phi(x, x_0, x_1, \dots, x_n)\}$ ”. sometimes instead of “ $\{x \mid x \in A \wedge \phi(x, x_0, x_1, \dots, x_n)\}$ ” we write “ $\{x \in A \mid \phi(x, x_0, x_1, \dots, x_n)\}$ ”.

Let’s talk about some axioms in set theoty. These daxioms will come in handy later.

Axiom 4.1.4. Existence there is a “Set” with no member i.e..

$$\exists x \quad Type_0(x)$$

definition 4.1.2. Empty set also known as the null set, is the unique mathematical set con-

taining no elements at all. (uniqueness can be proven.)

$$\emptyset := \{x \mid \perp\}$$

therefore.

$$\forall x(x \in \emptyset \leftrightarrow \perp \wedge Type_0(x))$$

therefore.

$$\forall x(x \notin \emptyset)$$

Example 4.1.2. $\therefore \forall X(\emptyset \subseteq X)$

1	$x \in \emptyset$	<i>acp</i>
2	$\forall x(x \notin \emptyset)$	<i>def</i>
3	$x \notin \emptyset$	$\forall E, 2$
4	$\perp$	$\perp I, 1, 3$
5	$x \in X$	$\perp E, 4$
6	$\emptyset \subseteq X$	$\subseteq I, 1, 5$
7	$\forall X(\emptyset \subseteq X)$	$\forall I, 6$

Lemma 4.1.1. $A = \emptyset \therefore \therefore \forall y(y \notin A)$

1	$\forall y(y \notin A)$	<i>ass</i>
2	$y \in A$	<i>acp</i>
3	$y \notin A$	$\forall E, 1$
4	$\perp$	$\perp I, 2, 3$
5	$y \in \emptyset$	$\perp E, 4$
6	$y \in \emptyset$	<i>acp</i>
7	$\forall y(y \notin \emptyset)$	<i>def.</i>
8	$y \notin \emptyset$	$\forall E, 7$
9	$\perp$	$\perp I, 6, 8$
10	$y \in A$	$\perp E, 9$
11	$A = \emptyset$	$I =$

1	$A = \emptyset$	<i>ass</i>
2	$\forall y(y \notin \emptyset)$	<i>def.</i>
3	$\forall y(y \notin A)$	$= E, 1, 2$

Theorem 4.1.3. (*schema*) $(\emptyset \neq I, \emptyset \neq E, \emptyset = I, \emptyset = E)$

$\frac{a \in A}{\therefore A \neq \emptyset} \quad \emptyset \neq I$	$\begin{array}{c} A \neq \emptyset \\ \left \begin{array}{c} x \in A \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \emptyset \neq E$
$\begin{array}{c} \left \begin{array}{c} x \in A \\ \vdots \\ \perp \end{array} \right. \\ \therefore A = \emptyset \end{array} \quad \emptyset = I$	$\frac{A = \emptyset}{\therefore a \notin A} \quad \emptyset = E$

- 1 $a \in A$ *ass*
- 2 $\exists x(x \in A)$ $\exists I, 1$
- 3 $\neg(\forall x(x \notin A))$ *De Morgan* 2
- 4 $A \neq \emptyset$ *lemma 4.4.1, 3*

- 1 $\left| \begin{array}{c} x \in A \\ \vdots \\ \perp \end{array} \right.$ *acp*
- n $\left| \begin{array}{c} x \in A \\ \vdots \\ \perp \end{array} \right.$
- $n+1$ $x \notin A$ $\neg I$
- $n+2$ $\forall x(x \notin A)$ $\forall I, n+1$
- $n+3$ $A = \emptyset$ *lemma 4.4.1, 1*

- 1 $A = \emptyset$ *ass*
- 2 $\forall x(x \notin A)$ *lemma 4.4.1, 1*
- 3 $a \notin A$ $\forall E2$

Axiom 4.1.5. Existence

$$Type_0(\emptyset)$$

definition 4.1.3. Pairing set

$$\{a_0, a_1, a_2, \dots, a_n\} := \{x \mid x = a_0 \vee x = a_1 \vee x = a_2 \vee \dots \vee x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, \dots, a_n\} \leftrightarrow (Type_0(x) \wedge x = a_0 \vee x = a_1 \vee x = a_2 \vee \dots \vee x = a_n))$$

Theorem 4.1.4. (*schema*) $(\{ \} I, \{ \} E)$

$\frac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, \dots, a_n\}} \quad \{ \} I$	$a \in \{a_0, a_1, a_2, \dots, a_n\}$ $\left \begin{array}{l} a = a_0 \\ \vdots \\ \theta \end{array} \right.$ $\vdots$ $\left \begin{array}{l} a = a_n \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \quad \{ \} E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Axiom 4.1.6. Pairing (*schema*)

$$Type_0(\{a_0, a_1, a_2, \dots, a_n\})$$

definition 4.1.4. Union and Intersection

Intersection:

$$A \cap B := \{x \mid x \in A \wedge x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \wedge x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \vee x \in B\}$$

therefore.

$$\forall x(x \in A \cup B \leftrightarrow (x \in A \vee x \in B))$$

Theorem 4.1.5. (*schema*) ($\cap I, \cap E, \cup I, \cup E$)

$\frac{a \in A}{a \in B}$ $\frac{a \in B}{\therefore a \in A \cap B} \quad \cap I$	$\frac{a \in A \cap B}{\therefore a \in A} \quad \cap E$ $\frac{a \in A \cap B}{\therefore a \in B} \quad \cap E$
$\frac{a \in A}{\therefore a \in A \cup B} \quad \cup I$ $\frac{a \in A}{\therefore a \in B \cup A} \quad \cup I$	$a \in A \cup B$ $\left \begin{array}{l} a \in A \\ \vdots \\ \theta \end{array} \right.$ $\left \begin{array}{l} a \in B \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \quad \cup E$

Proof Is Left As An Exercise To The Reader. 😊

sometimes all members of a set is also set we call it “set of sets”.

definition 4.1.5. Big Union and Intersection if “ A ” is “set of sets” and “ $A \neq \emptyset$ ” we define Big Intersection:

$$\bigcap A := \{x \mid \forall X (X \in A \rightarrow x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x \left(x \in \bigcap A \leftrightarrow (Type_0(x) \wedge \forall X (X \in A \rightarrow x \in X)) \right)$$

Big Union:

$$\bigcup A := \{x \mid \exists X (X \in A \wedge x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x \left(x \in \bigcup A \leftrightarrow (Type_0(x) \wedge \exists X (X \in A \wedge x \in X)) \right)$$

Theorem 4.1.6. (*schema*) $(\cap I, \cap E, \cup I, \cup E)$

$\begin{array}{c} \left \begin{array}{l} X \in A \\ \vdots \\ a \in X \end{array} \right. \\ \hline \therefore a \in \bigcap A \quad \bigcap I \end{array}$	$\begin{array}{c} a \in \bigcap A \\ B \in A \\ \hline \therefore a \in B \quad \bigcap E \end{array}$
$\begin{array}{c} a \in B \\ B \in A \\ \hline \therefore a \in \bigcup A \quad \bigcup I \end{array}$	$\begin{array}{c} a \in \bigcup A \\ \left \begin{array}{l} a \in X \\ X \in A \\ \vdots \\ \theta \end{array} \right. \\ \hline \therefore \theta \quad \bigcup E \end{array}$

Proof Is Left As An Exercise To The Reader. 😊

definition 4.1.6. Complement

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow \text{Type}_0(x) \wedge x \notin A)$$

Theorem 4.1.7. (schema) $((\)^c I, (\)^c E)$

$\begin{array}{c} \left \begin{array}{l} a \in A \\ \vdots \\ \perp \end{array} \right. \\ \hline \therefore a \notin A \quad (\)^c I \end{array}$	$\begin{array}{c} \left \begin{array}{l} a \notin A \\ \vdots \\ \perp \end{array} \right. \\ \hline \therefore a \in A \quad (\)^c E \end{array}$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Observation 4.1.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow \text{Type}_0(x) \wedge \top)$$

therefore.

$$\forall x(x \in V)$$

definition 4.1.7. Subtraction

$$A - B := \{x \mid x \in A \wedge x \notin B\}$$

therefore.

$$\forall x \left(x \in A - B \leftrightarrow (Type_0(x) \wedge x \in A \wedge x \notin B) \right)$$

Theorem 4.1.8. (schema) $(-I, -E)$

$\frac{a \in A \quad a \notin B}{\therefore a \in A - B} \quad -I$	$\frac{a \in A - B}{\therefore a \in A} \quad -E \quad \frac{a \in A - B}{\therefore a \notin B} \quad -E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Lemma 4.1.2. $\therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader. 😊

Axiom 4.1.7. Seperation

$$\forall X, Y \quad Type_0(X) \rightarrow Type_0(X \cap Y)$$

therefore.

$$Type_0(A) \rightarrow Type_0(A \cap B)$$

therefore.

$$Type_0\left(\bigcap A\right)$$

therefore.

$$Type_0(A) \rightarrow Type_0(A - B)$$

Axiom 4.1.8. union

$$\forall X \quad Type_0(X) \rightarrow Type_0\left(\bigcup X\right)$$

therefore.

$$Type_0(A) \wedge Type_0(B) \rightarrow Type_0(A \cup B)$$

definition 4.1.8. Powerset

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X \left(X \in \mathcal{P}(A) \leftrightarrow (Type_0(X) \wedge X \subseteq A) \right)$$

Theorem 4.1.9. (schema) $(\mathcal{P}I, \mathcal{P}E)$

$\frac{A \subseteq B}{\therefore A \in \mathcal{P}(B)} \quad \mathcal{P}I$	$\frac{A \in \mathcal{P}(B)}{\therefore A \subseteq B} \quad \mathcal{P}E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Axiom 4.1.9. Powerset

$$\forall X \quad \text{Type}_0(X) \rightarrow \text{Type}_0(P(X))$$

from now on we don't get in to all details of proofs because it will be too long and hard to read. so we will take it easy from now and don't write all details like reasons of lines and all lines we left it to reader to undrestand what happened there. 😊

Chapter 5

Relations and Functions

5.1 Orderd pair and Cartesian product

definition 5.1.1. *Ordered pair*

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

so $\forall x, y \quad \text{Type}_0((x, y))$

definition 5.1.2. *Tuples*(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)

$$(x_1, x_2, \dots, x_n) := ((x_1, x_2, \dots, x_{n-1}), x_n)$$

Lemma 5.1.1. $\therefore (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader. 😊

definition 5.1.3. *Cartesian product*

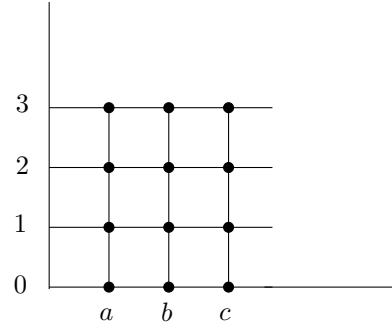
$$A \times B := \{u \mid \exists x, y \left((x, y) = u \wedge x \in A \wedge y \in B \right)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 5.1.1. if $A = \{a, b, c\}$ and $B = \{0, 1, 2, 3\}$ then

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



so $\forall X, Y \quad Type_0(X) \wedge Type_0(Y) \rightarrow Type_0(X \times Y)$

Theorem 5.1.1. (schema) $(\times I, \times E)$

$\frac{t \in A \quad s \in B}{\therefore (t, s) \in A \times B} \quad \times I$	$\begin{array}{l} \therefore t \in A \times B \\ \left \begin{array}{l} t = (x, y) \\ x \in A \\ y \in B \\ \vdots \\ \theta \end{array} \right. \\ \theta \qquad \qquad \times E \end{array}$
---	---

Proof Is Left As An Exercise To The Reader. 😊

Theorem 5.1.2. (schema)

$\frac{(t, s) \in A \times B}{\therefore t \in A}$	$\frac{(t, s) \in A \times B}{\therefore s \in B}$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Exercise 5.1.1. for sets A, B, C , prove.

1. $\therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$
2. $\therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$
3. $\therefore A \times (B - C) = (A \times B) - (A \times C)$

5.2 Relations

definition 5.2.1. Relation we call a set like R a **Relation** when there are two sets like X and Y s.t. $R \subseteq X \times Y$.

$$Rel(R) : \exists X, Y (R \subseteq X \times Y).$$

sometimes we use different notations in this book. you have to be smart and recognize it by yourself. sometimes instead of " $x \in R$ " we denote it by " $R(x)$ ". and instead of " $(x_0, x_1, \dots, x_n) \in R$ " we denote it by " $R(x_0, x_1, \dots, x_n)$ ". if tuple is binary, instead of " $(x, y) \in R$ " we denote it by " xRy ". or " $R(x, y)$ " instead of " $R((x, y))$ "

so if we have a definition like " $R(x) : \phi(x)$ " s.t. " $\phi(x)$ " is a first order proposition. "weak comprehension" show us there is a straight relation between "propositions" and "sets" so you can see it as " $R := \{x \mid \phi(x)\}$ ". and sometimes we might show a formula in natural language for example below instead of " $Rel(X)$ " we can say " X is relation". if it is confusing so forget it. 😊

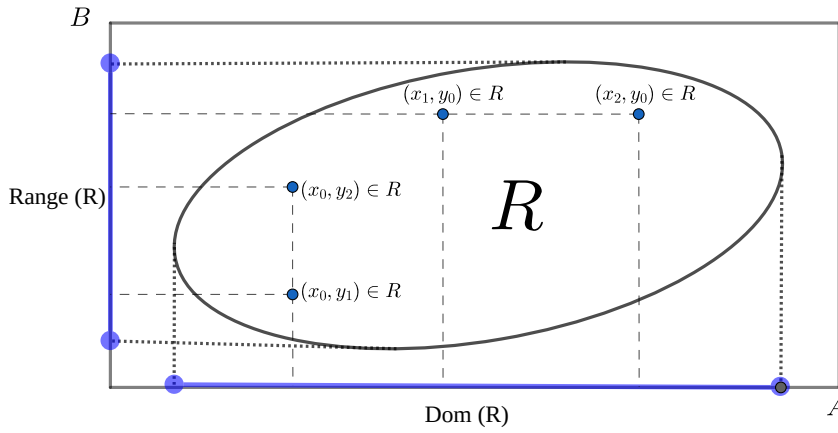
definition 5.2.2. Domain and Range(also called Image)

Domain of a set " R " is all " x " that there exists a " y " s.t. " xRy "

$$Dom(R) := \{x \mid \exists y \ xRy\}$$

Range(Image) of a set " R " is all " y " that there exists a " x " s.t. " xRy "

$$Range(R) := \{y \mid \exists x \ xRy\}.$$



Theorem 5.2.1. (schema) $(DomI, DomE)$

$\frac{tRs}{\therefore t \in \text{Dom}(R)} \quad \text{Dom}I$	$\begin{array}{c} t \in \text{Dom}(R) \\ \left \begin{array}{c} tRy \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \text{Dom}E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Theorem 5.2.2. (schema) ($\text{Range}I, \text{Range}E$)

$\frac{tRs}{\therefore s \in \text{Range}(R)} \quad \text{Range}I$	$\begin{array}{c} s \in \text{Range}(R) \\ \left \begin{array}{c} xRs \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \text{Range}E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

definition 5.2.3. *Inverse of a Relation*

$$R^{-1} = \{u \mid \exists x, y (u = (x, y) \wedge yRx)\}$$

or i.e.

$$R^{-1} = \{(x, y) \mid yRx\}$$

therefore.

$$\forall x, y (xR^{-1}y \leftrightarrow yRx)$$

Theorem 5.2.3. (schema) ($()^{-1}I, ()^{-1}E$)

$\frac{tRs}{\therefore sR^{-1}t} \quad ()^{-1}I$	$\begin{array}{c} t \in R^{-1} \\ \left \begin{array}{c} t = (x, y) \\ yR^{-1}x \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad ()^{-1}E$
--	---

Proof Is Left As An Exercise To The Reader. 😊

definition 5.2.4. *Combination of a Relations*

$$S \circ R = \{u \mid \exists x, y (u = (x, y) \wedge \exists z (xRz \wedge zSy))\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z (xRz \wedge zSy)\}$$

therefore.

$$\forall x, y ((x, y) \in S \circ R \leftrightarrow \exists z (xRz \wedge zSy))$$

Theorem 5.2.4. (schema) $(\circ I, \circ E)$

$\frac{tSs \quad sRr}{\therefore sR \circ Sr} \quad \circ I$	$\begin{array}{l} t \in R \circ S \\ \left \begin{array}{l} t = (x, y) \\ xSz \\ zRy \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \circ E$
--	---

Proof Is Left As An Exercise To The Reader. 😊

definition 5.2.5. *Image*

$$R[A] := \{y \mid \exists x \in A \quad xRy\}$$

you can also see by definition of “inverse of a Relation” and “image” this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad xRy\}$$

we cal this “**Pre-image**”

Theorem 5.2.5. (schema) $(R \sqcap I, R \sqcap E, R^{-1} \sqcap I, R^{-1} \sqcap E)$

$\frac{t \in A}{\therefore s \in R[A]} \quad R \sqcap I$	$\frac{s \in R[A] \quad \left \begin{array}{l} x \in A \\ xRs \\ \vdots \\ \theta \end{array} \right.}{\therefore \theta} \quad R \sqcap E$
$\frac{s \in B}{\therefore t \in R^{-1}[B]} \quad R^{-1} \sqcap I$	$\frac{t \in R^{-1}[B] \quad \left \begin{array}{l} y \in B \\ tRy \\ \vdots \\ \theta \end{array} \right.}{\therefore \theta} \quad R^{-1} \sqcap E$

Proof Is Left As An Exercise To The Reader. 😊

5.3 Functions

definition 5.3.1. *Function* we call a relation “**f**” a “**function**” when every input has exactly one output i.e.

$$Func(f) : Rel(f) \wedge \forall x, y_1, y_2 \left((x, y_1) \in f \wedge (x, y_2) \in f \rightarrow y_1 = y_2 \right)$$

Theorem 5.3.1. (schema) $(FuncI, FuncE)$

$\frac{\left \begin{array}{l} xfy \\ x fz \\ \vdots \\ y = z \end{array} \right.}{\therefore Func(f)} \quad FuncI$	$\frac{Func(f) \quad tfs \quad tfr}{\therefore s = r} \quad FuncE$
---	--

Proof Is Left As An Exercise To The Reader. 😊

definition 5.3.2. *Restriction of a function* to a set “**X**” (usually a subset of “ $dom(f)$ ”) is the function

$$f \upharpoonright X := \{(x, y) \in f \mid x \in X\}$$

definition 5.3.3. Output of a function assume a function f and $(x, y) \in f$, we define “ $f(x) := y$ ” and we call “ $f(x)$ ” a “**output of a function**”. and the reverse is also true, if we use “ $f(x)$ ”, we have a function “ f ” and we can say “ $x \in \text{Dom}(f)$ ”. “output of a function” is object too.

Theorem 5.3.2. (schema) $(\forall I, \forall E, \exists I, \exists E)$

$\frac{\text{Func}(f)}{(x, y) \in f} \quad \frac{}{\therefore f(x) = y} \quad f()E$	$\frac{f(x) = y}{\text{Func}(f)} \quad \frac{}{\therefore (x, y) \in f} \quad f()E$
---	---

Attention 5.3.1. input of a function can be a ordered pair itself, so we can have $f((x, y))$ and it is a “**output of a function**”. we define “ $f(x, y) := f((x, y))$ ” for simplicity. this is true for tuples as well, so we can have “ $f((x_1, x_2, \dots, x_n))$ ” and we define “ $f(x_1, x_2, \dots, x_n) := f((x_1, x_2, \dots, x_n))$ ” for simplicity.

definition 5.3.4. Term we define term like this.

1. every obj is a term.
2. if $t_0, \dots, t_n$ are terms and “ $\text{Func}(f)$ ” and “ $(t_0, \dots, t_n) \in \text{Dom}(f)$ ”. then “ $f(t_0, \dots, t_n)$ ” is also a term.

we denote terms with small letters like $t, s, r, \dots$

definition 5.3.5. Substitution for term we define Substitution an term with variable like this.

1. for variables if $x = y$ then $y[t/x] := t$ and if $x \neq y$ then $y[t/x] := y$.
2. for constants $c[t/x] := c$.
3. for terms like $f(t_0, \dots, t_n)$

$$f(t_0, \dots, t_n)[t/x] := f(t_0[t/x], \dots, t_n[t/x])$$

definition 5.3.6. Substitution for formulas we define Substitution for a formula like this.

1. for atomic formulas we define it like this. if it's a single term we do it as we explain in “**Substitution for term**”. if it's like $t_0 = t_1$ or $t_0 \in t_1$ (t_0, t_1 are terms). we define it like this.

$$\text{Set}(y)[t/x] := \text{Set}(y[t/x])$$

$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$

$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. if formula A is complex we define it like this.

$$(B \circ C)[t/x] := (B[t/x] \circ C[t/x])$$

($\circ$ can be $\rightarrow, \wedge, \vee$).

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 5.3.3. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$

Rules	Terms and Conditions
$\frac{\phi(x)}{\therefore \forall x \phi(x)} \quad \forall I$	1. x is a variable. 2. "$\phi(x)$" must not be assumed.
$\frac{\forall x \phi(x)}{\therefore \phi[t/x]} \quad \forall E$	1. t is a term.
$\frac{\phi[t/x]}{\therefore \exists x \phi(x)} \quad \exists I$	1. t is a term.
$\begin{array}{c} \exists x \phi(x) \\ \left \begin{array}{c} \phi(x) \\ \vdots \\ \theta \end{array} \right. \\ \therefore \theta \end{array} \quad \exists E$	1. x is a variable. 2. "$\phi(x)$" must not be assumed in previous lines. (except ones that are closed) 3. x is not free in θ

$(\forall I, \exists E)$ is the same as $(\forall I, \exists E)$ in chapter three so we have to prove $(\forall E, \exists I)$. so assume $t = f(x)$ so to it's easy to see $\text{Func}(f)$ and $x \in \text{Dom}(f)$.

1	$\forall x \phi(x)$	ass	1	$\phi(f(x))$	ass
2	$\text{Func}(f)$	1	2	$\text{Func}(f)$	ass
3	$x \in \text{Dom}(f)$	1	3	$x \in \text{Dom}(f)$	ass
4	$(x, y) \in f$	acp	4	$(x, y) \in f$	acp
5	$f(x) = y$	def	5	$f(x) = y$	def
6	$\phi(y)$	$\forall E$	6	$\phi(y)$	$= E$
7	$\phi(f(x))$	$= E$	7	$\exists x \phi(x)$	$\exists I$
8	$\phi(f(x))$	$\exists E$	8	$\exists x \phi(x)$	$\exists E$

Attention 5.3.2. in all rules and theorems you can replace all constants like $a, b, c, \dots$ with terms like $t, s, r, \dots$ and rewrite them.

definition 5.3.7. Compatible Function

1. $\text{Compatible}(f, g) : \forall x \in \text{dom}(f) \cap \text{dom}(g) \quad f(x) = g(x)$.
2. $\text{Compatible}(F) : \forall f, g \in F \quad \text{Compatible}(f, g)$.

Lemma 5.3.1. .

1. $Compatible(f, g) \therefore / \therefore Func(f \cup g)$.
2. $Compatible(f, g) \therefore / \therefore f \upharpoonright (dom(f) \cap dom(g)) = g \upharpoonright (dom(f) \cap dom(g))$.

Proof Is Left As An Exercise To The Reader. 😊

Theorem 5.3.4. $Compatible(F) / \therefore (1), (2), (3)$

1. $Func(\bigcup F)$.
2. $dom(\bigcup F) = \bigcup \{dom(f) \mid f \in F\}$.
3. $\forall f \in F \quad Compatible(f, \bigcup F)$.

Proof Is Left As An Exercise To The Reader. 😊

definition 5.3.8. Function from set to set we call a function f from A to B when it's domain is A and it's range is subset of B i.e.

$$(f : A \rightarrow B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 5.3.9. Set of functions from set to set we denote set of all " $f : A \rightarrow B$ " with " B^A " i.e.

$$B^A : \{f \mid f : A \rightarrow B\}$$

definition 5.3.10. Injective function(Relation) we call a function(Relation) f injective when every output has at most one input i.e.

$$Injective(f) : \forall y, x_1, x_2 \left((x_1, y) \in f \wedge (x_2, y) \in f \rightarrow x_1 = x_2 \right).$$

we call a function f injective from A to B when it's function from A to B and it's injective i.e.

$$(f : A \xrightarrow{inj} B) : (f : A \rightarrow B) \wedge Injective(f)$$

Theorem 5.3.5. (schema) $(InjectiveI, InjectiveE)$

$\begin{array}{l} x f z \\ y f z \\ \vdots \\ x = y \end{array}$ $\therefore Injective(f) \quad InjectiveI$	$\begin{array}{l} Injective(f) \\ t f r \\ \hline s f r \\ \therefore s = r \end{array} \quad InjectiveE$
---	---

Proof Is Left As An Exercise To The Reader. 😊

definition 5.3.11. Surjective function we call a function f from A to B when it's domain is A and it's range is equal to B i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \rightarrow B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[\text{surj}]{} B) : (f : A \rightarrow B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 5.3.12. Bijection function we call a function f from A to B when it's domain is A and it's range is equal to B i.e.

$$(f : A \xrightarrow[\text{surj}]{\text{inj}} B) : (f : A \xrightarrow{\text{inj}} B) \wedge (f : A \xrightarrow[\text{surj}]{} B)$$

5.4 Family of sets

Attention 5.4.1. Indexes are objects (like numbers) allows us to give a unique "tag" to every object. we use $i, j, k, \dots$ for variable indexes and we use $m, n, l, \dots$ for constant indexes.

definition 5.4.1. Family of sets assume $A : I \rightarrow \mathcal{F}$ is a function from a nonempty set of indices I to a family of sets $\mathcal{F}$, then we call $(\text{Range}[A])$ an indexed family of sets. we can write A_i instead of $A(i)$ and we can write $(A_i)_{i \in I}$ instead of " A " and $\{A_i\}_{i \in I}$ instead of " $\text{Range}[A]$ " or " $A[I]$ ".

$$(A_i)_{i \in I} := A$$

and

$$\{A_i\}_{i \in I} := A[I]$$

definition 5.4.2. Intersrction and union of an family of sets

$$\bigcap_{i \in I} A_i := \bigcap \{A_i\}_{i \in I}$$

$$\bigcup_{i \in I} A_i := \bigcup \{A_i\}_{i \in I}$$

Attention 5.4.2. we can write $\forall i \in I \quad P(i)$ instead of $\forall i (i \in I \rightarrow P(i))$ and we can write $\exists i \in I \quad P(i)$ instead of $\exists i (i \in I \wedge P(i))$.

Lemma 5.4.1.

$$/ \therefore \bigcap_{i \in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i \in I} A_i$	<i>acp</i>
2	$x \in \bigcap \{A_i\}_{i \in I}$	<i>def.1</i>
3	$x \in \bigcap A[I]$	<i>def.1</i>
4	$\forall Y \in A[I] \quad x \in Y$	<i>def</i> $\bigcap$
5	$\forall Y ((\exists i \in I \quad A_i = Y) \rightarrow x \in Y)$	<i>def image</i>
6	$(\exists i \in I \quad A_i = A_{i'}) \rightarrow x \in A_{i'}$	$\forall E$
7	$i' \in I$	<i>acp</i>
8	$A_{i'} = A_{i'}$	$= I$
9	$i' \in I \wedge A_{i'} = A_{i'}$	$\wedge I$
10	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
11	$x \in A_{i'}$	$\rightarrow E$
12	$i' \in I \rightarrow x \in A_{i'}$	$\rightarrow I$
13	$\forall i \in I \quad x \in A_i$	$\forall I$
14	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	<i>def</i>
15	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	<i>acp</i>
16	$\forall i \in I \quad x \in A_i$	<i>def</i>
17	$\exists i \in I \quad A_i = Y$	<i>acp</i>
18	$i \in I$	<i>acp</i>
19	$x \in A_i$	$\forall E$
20	$A_i = Y$	$\wedge E$
21	$x \in Y$	$= E$
22	$x \in Y$	$\exists E$
23	$(\exists i \in I \quad A_i = Y) \rightarrow x \in Y$	$\rightarrow I$
24	$\forall Y ((\exists i \in I \quad A_i = Y) \rightarrow x \in Y)$	$\forall I$
25	$\forall Y \in A[I] \quad x \in Y$	<i>def</i>
26	$x \in \bigcap A[I]$	<i>def</i>
27	$x \in \bigcap \{A_i\}_{i \in I}$	<i>def</i>
28	$x \in \bigcap_{i \in I} A_i$	<i>def</i>
29	$\bigcap_{i \in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/ \therefore \bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$x \in \bigcup_{i \in I} A_i$	<i>acp</i>
2	$x \in \bigcup \{A_i\}_{i \in I}$	<i>def.1</i>
3	$x \in \bigcup A[I]$	<i>def.1</i>
4	$\exists Y \in A[I] \quad x \in Y$	<i>def</i> $\bigcup$
5	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	<i>def.image</i>
6	$x \in Y$	<i>acp</i>
7	$\exists i \in I \quad A_i = Y$	<i>acp</i>
8	$i \in I$	<i>acp</i>
9	$A_i = Y$	<i>acp</i>
10	$x \in A_i$	$= E$
11	$\exists i \in I \quad x \in A_i$	$\exists I$
12	$\exists i \in I \quad x \in A_i$	$\exists E$
13	$\exists i \in I \quad x \in A_i$	$\exists E$
14	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	<i>def</i>
15	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	<i>acp</i>
16	$\exists i \in I \quad x \in A_i$	<i>def</i>
17	$i' \in I$	<i>acp</i>
18	$x \in A_{i'}$	<i>acp</i>
19	$A_{i'} = A_{i'}$	$= I$
20	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
21	$(\exists i \in I \quad A_i = A_{i'}) \wedge x \in A_{i'}$	$\wedge I$
22	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists I$
23	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists E$
24	$\exists Y \in A[I] \quad x \in Y$	<i>def</i>
25	$x \in \bigcup A[I]$	<i>def</i>
26	$x \in \bigcup \{A_i\}_{i \in I}$	<i>def</i>
27	$x \in \bigcup_{i \in I} A_i$	<i>def</i>
28	$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$= I$

Theorem 5.4.1. (*schema*) $(\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E)$

$\begin{array}{c} \left \begin{array}{l} i \in I \\ \vdots \\ t \in A_i \end{array} \right. \\ \hline \therefore t \in \bigcap_{i \in I} A_i \quad \bigcap_I I \end{array}$	$\begin{array}{c} t \in \bigcap_{i \in I} A_i \\ \hline n \in I \\ \hline \therefore t \in A_n \quad \bigcap_I E \end{array}$
$\begin{array}{c} t \in A_n \\ \hline n \in I \\ \hline \therefore t \in \bigcup_{i \in I} A_i \quad \bigcup_I I \end{array}$	$\begin{array}{c} t \in \bigcup_{i \in I} A_i \\ \left \begin{array}{l} i \in I \\ t \in A_i \\ \vdots \\ \theta \end{array} \right. \\ \hline \therefore \theta \quad \bigcup_I E \end{array}$

Proof Is Left As An Exercise To The Reader. 😊

5.5 Structures and morphisms

definition 5.5.1. *Structure “ $\mathfrak{M}$ ” is a tuple defined like this.*

$$\mathfrak{M} := (M, f_0, f_1, \dots, f_n, R_0, R_1, \dots, R_m, c_0, c_1, \dots, c_k)$$

“ M ” is a **nonempty** set shows our universe “ $\forall i \in \{0, \dots, n\} \quad f_i : M^{n'} \rightarrow M$ ” are functions and “ $\forall i \in \{0, \dots, m\} \quad R_i \subseteq M^{m'}$ ” are relations and “ $\forall i \in \{0, \dots, k\} \quad c_i \in M$ ” are constants.

(M^n means $\overbrace{M \times M \times \dots \times M}^{n \text{ times}}$ we will define it later) and for the case we have arbitrary family of functions and relations and constants.

$$\mathfrak{M} := (M, \{f_i\}_{i \in I_f}, \{R_i\}_{i \in I_R}, \{c_i\}_{i \in I})$$

definition 5.5.2. Homomorphism if “ $\mathfrak{M} := (M, \{f_{i,m}\}_{i \in I_f}, \{R_{i,m}\}_{i \in I_R}, \{c_{i,m}\}_{i \in I})$ ” and “ $\mathfrak{N} := (N, \{f_{i,n}\}_{i \in I_f}, \{R_{i,n}\}_{i \in I_R}, \{c_{i,n}\}_{i \in I})$ ” are two structures. function “ $h : M \rightarrow N$ ” is called “(weak) Homomorphism” when this conditions holds

1. for every relation “ $R_{i,m}$ ”

$$\forall x_0, x_1, x_2, \dots, x_k \in M \quad R_{i,m}(x_0, x_1, x_2, \dots, x_k) \rightarrow R_{i,n}(h(x_0), h(x_1), h(x_2), \dots, h(x_k))$$

2. for every function “ $f_{i,m}$ ”

$$\forall x_0, x_1, x_2, \dots, x_k \in M \quad h(f_{i,m}(x_0, x_1, x_2, \dots, x_k)) = f_{i,n}(h(x_0), h(x_1), h(x_2), \dots, h(x_k))$$

3. for every constant " $c_{i,m}$ "

$$h(c_{i,m}) = c_{i,n}$$

in condition (1) if " $\leftrightarrow$ " be instead of " $\rightarrow$ " then it's called "(strong) Homomorphism"

few notices: when we are talking about "Homomorphism" we usually talk about "(strong) Homomorphism".

in the case (in definition "(strong) Homomorphism") we denote it by " $Hom(h, \mathfrak{M}, \mathfrak{N})$ ". if exists function " h " " $Hom(h, \mathfrak{M}, \mathfrak{N})$ " we say " $\mathfrak{M}$ " and " $\mathfrak{N}$ " are homomorphic. we denote it by " $Hom(\mathfrak{M}, \mathfrak{N})$ ".

if " $h : M \xrightarrow{inj} N$ " we call it "Monomorphism" or "Embedding".we say " $\mathfrak{M}$ " and " $\mathfrak{N}$ " are momomorphic.

if " $h : M \xrightarrow[surj]{} N$ " we call it "Epimorphism".we say " $\mathfrak{M}$ " and " $\mathfrak{N}$ " are Eomomorphic.

if " $h : M \xrightarrow[surj]{inj} N$ " we call it "Isomorphism".we say " $\mathfrak{M}$ " and " $\mathfrak{N}$ " are Iomomorphic. and denote it by " $\mathfrak{M} \cong \mathfrak{N}$ ".

if " $k : M \rightarrow M$ " is a function and " $Hom(k, \mathfrak{M}, \mathfrak{M})$ ". we call " k " "Indomorphism".

if " $k : M \xrightarrow[surj]{inj} M$ " is a bijection and " $Hom(k, \mathfrak{M}, \mathfrak{M})$ ". we call " k " "Automorphism".

5.6 Equivalences and Partitions

definition 5.6.1. Equivalence relation assume a structure " $\mathfrak{A} := (A, \sim)$ " (" $\sim$ " is a relation). we say " $\sim$ " is a "Equivalence relation" in " $\mathfrak{A}$ " if it's reflexive, symmetric, and transitive.

$$(reflexive) : \forall x \in A \quad x \sim x$$

$$(symmetric) : \forall x, y \in A \quad x \sim y \rightarrow y \sim x$$

$$(transitive) : \forall x, y, z \in A \quad x \sim y \wedge y \sim z \rightarrow x \sim z$$

Lemma 5.6.1. " $\cong$ " is a equivalence relation in " $(Structure, \cong)$ "

Proof Is Left As An Exercise To The Reader. 😊

definition 5.6.2. Equivalence class of " x " Let " $\sim$ " be an equivalence relation on " $\mathfrak{A}$ ". For every " $x \in A$ "

$$[x]_{\sim} := \{y \in A \mid y \sim x\}$$

you can also see it as a function $[_]_{\sim} : A \rightarrow V$.

Theorem 5.6.1. (schema) ($[_]_{\sim} I, [_]_{\sim} E$)

$\frac{t \sim s}{\therefore t \in [s]_{\sim}} \quad [_]_{\sim} I$	$\frac{t \in [s]_{\sim}}{\therefore t \sim s} \quad [_]_{\sim} E$
--	--

Proof Is Left As An Exercise To The Reader. 😊

Lemma 5.6.2. *For every “ $a, b \in A$ ”*

1. $a \sim b \therefore / \therefore [a]_{\sim} = [b]_{\sim}$
2. $a \not\sim b \therefore / \therefore [a]_{\sim} \cap [b]_{\sim} = \emptyset$

prove.1

			$/ \therefore$		
1	$a \sim b$	ass			
2	$x \in [a]_{\sim}$	acp			
3	$x \sim a$	$[\sim]E$			$\therefore /$
4	$x \sim b$	$trans$	1	$[a]_{\sim} = [b]_{\sim}$	ass
5	$x \in [b]_{\sim}$	$[\sim]I$	2	$a \sim a$	$reflex$
6	$x \in [b]_{\sim}$	acp	3	$a \in [a]_{\sim}$	$[\sim]I$
7	$x \sim b$	$[\sim]E$	4	$a \in [b]_{\sim}$	$= E$
8	$x \sim a$	$trans$	5	$a \sim b$	$[\sim]E$
9	$x \in [a]_{\sim}$	$[\sim]I$			
10	$[a]_{\sim} = [b]_{\sim}$				

prove.2

			$/ \therefore$		
1	$a \not\sim b$	ass			
2	$[a]_{\sim} \cap [b]_{\sim} \neq \emptyset$	acp			$\therefore /$
3	$x \in [a]_{\sim} \cap [b]_{\sim}$	acp	1	$[a]_{\sim} \cap [b]_{\sim} = \emptyset$	
4	$x \in [a]_{\sim}$	$\cap E$	2	$a \sim b$	acp
5	$x \in [b]_{\sim}$	$\cap E$	3	$a \sim a$	$reflex$
6	$x \sim a$	$[\sim]E$	4	$a \in [a]_{\sim}$	$[\sim]I$
7	$x \sim b$	$[\sim]E$	5	$a \in [b]_{\sim}$	$[\sim]I$
8	$a \sim b$	$trans$	6	$a \in [a]_{\sim} \cap [b]_{\sim}$	$\cap I$
9	$\perp$		7	$\perp$	
10	$\perp$		8	$a \not\sim b$	
11	$[a]_{\sim} \cap [b]_{\sim} = \emptyset$				

definition 5.6.3. *Partition* is a “set of sets” like “ P ” of a set like “ A ” with this conditions.

1. $\forall X \in P \quad X \neq \emptyset$
2. $\forall X, Y \in P \quad X \neq Y \rightarrow X \cap Y = \emptyset$

3. $\bigcup P = A$
 if these condition holds then we say “ $\text{Part}(P, A)$ ”.

definition 5.6.4. Set of Equivalence classes

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

Theorem 5.6.2. if “ $\sim$ ” is a equivalence class in “ $(A, \sim)$ ” then “ $A/\sim$ ” is a partition for “ A ” i.e. “ $\text{Part}(A/\sim, A)$ ”.

1	$[x]_\sim \in A/\sim$	<i>acp</i>
2	$x \sim x$	<i>reflex</i>
3	$x \in [x]_\sim$	$\square_\sim I$
4	$[x]_\sim \neq \emptyset$	$\neq \emptyset I$
5	$\forall [x]_\sim \in A/\sim \quad [x]_\sim \neq \emptyset$	
1	$[x]_\sim \neq [y]_\sim$	<i>acp</i>
2	$x \approx y$	<i>Lemma 5.6.2.1</i>
3	$[x]_\sim \cap [y]_\sim = \emptyset$	<i>Lemma 5.6.2.2</i>
4	$\forall [x]_\sim, [y]_\sim \in A/\sim \quad [x]_\sim \neq [y]_\sim \rightarrow [x]_\sim \cap [y]_\sim = \emptyset$	
1	$x \in \bigcup (A/\sim)$	<i>acp</i>
2	$[y]_\sim \in A/\sim$	<i>acp</i>
3	$x \in [y]_\sim$	<i>acp</i>
4	$[y]_\sim \subseteq A$	
5	$x \in A$	$\subseteq E$
6	$x \in A$	$\bigcup E$
7	$x \in A$	
8	$x \in [x]_\sim$	
9	$[x]_\sim \in (A/\sim)$	
10	$x \in \bigcup (A/\sim)$	$\bigcup I$
11	$A = \bigcup (A/\sim)$	$= I$

Theorem 5.6.3. if “ P ” is a partition for “ A ” we define

$$\sim_P := \{(x, y) \mid \exists Y \in P \quad (x \in Y \wedge y \in Y)\}$$

then “ $\sim_P$ ” is a equivalence class in “ $(A, \sim_P)$ ”.

1	$x \in A$	<i>acp</i>
2	$x \in \bigcup P$	<i>Def</i>
3	$x \in Y$	<i>acp</i>
4	$Y \in P$	<i>acp</i>
5	$\exists Y \in P \quad (x \in Y \wedge x \in Y)$	
6	$x \sim_P x$	
7	$x \sim_P x$	
8	$\forall x \in A \quad x \sim_P x$	

1	$x, y \in A$	<i>acp</i>
2	$x \sim_P y$	<i>acp</i>
3	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$	<i>Def</i>
4	$Y \in P \wedge x \in Y \wedge y \in Y$	<i>acp</i>
5	$Y \in P \wedge y \in Y \wedge x \in Y$	<i>commutative</i> $\wedge$
6	$\exists Y \in P \quad (y \in Y \wedge x \in Y)$	
7	$y \sim_P x$	
8	$y \sim_P x$	
9	$x \sim_P y \rightarrow y \sim_P x$	
10	$\forall x, y \in A \quad x \sim_P y \rightarrow y \sim_P x$	

1	$x, y, z \in A$	<i>acp</i>
2	$x \sim_P y$	<i>acp</i>
3	$y \sim_P z$	<i>acp</i>
4	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$	
5	$\exists Y \in P \quad (y \in Y \wedge z \in Y)$	
6	$Y \in P \wedge x \in Y \wedge y \in Y$	
7	$Y' \in P \wedge y \in Y' \wedge z \in Y'$	
8	$Y \cap Y' \neq \emptyset$	
9	$Y = Y'$	
10	$Y \in P \wedge x \in Y \wedge z \in Y$	
11	$\exists Y \in P \quad (x \in Y \wedge z \in Y)$	
12	$x \sim_P z$	
13	$x \sim_P z$	
14	$x \sim_P z$	
15	$x \sim_P y \wedge y \sim_P z \rightarrow x \sim_P z$	
16	$\forall x, y, z \in A \quad x \sim_P y \wedge y \sim_P z \rightarrow x \sim_P z$	

5.7 Orderings

definition 5.7.1. (partial) Order relation assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a relation). we say “ $\preceq$ ” is a “(partial) Order relation” in “ $\mathfrak{A}$ ” if it’s reflexive, antisymmetric, and transitive.

$$(\text{reflexive}) : \forall x \in A \quad x \preceq x$$

$$(\text{antisymmetric}) : \forall x, y \in A \quad x \preceq y \wedge y \preceq x \rightarrow x = y$$

$$(\text{transitive}) : \forall x, y, z \in A \quad x \preceq y \wedge y \preceq z \rightarrow x \preceq z$$

definition 5.7.2. (partial) Strict order relation assume a structure “ $\mathfrak{A} := (A, \prec)$ ” (“ $\prec$ ” is a relation). we say “ $\prec$ ” is a “(partial) Strict order relation” in “ $\mathfrak{A}$ ” if it’s irreflexive, and transitive.

$$(\text{irreflexive}) : \forall x \in A \quad x \not\prec x$$

$$(\text{transitive}) : \forall x, y, z \in A \quad x \prec y \wedge y \prec z \rightarrow x \prec z$$

by two conditions bellow we can proof asymmetric too.

$$(\text{asymmetric}) : \forall x, y \in A \quad x \prec y \rightarrow y \not\prec x$$

now we will show you relation between “(partial) Order relation” and “(partial) Strict order relation”.

Theorem 5.7.1. .

1. if “ $\preceq$ ” is a “(partial) Order relation” in “ $\mathfrak{A}$ ” define “ $a \prec b : a \preceq b \wedge a \neq b$ ” prove that “ $\prec$ ” is “(partial) Strict order relation”.

1	$x \in A$	acp
2	$x \prec x$	acp
3	$x \neq x$	$\wedge E$
4	$x = x$	$= I$
5	$\perp$	$\perp I$
6	$x \not\prec x$	$\neg I$
7	$\forall x \in A \quad x \not\prec x$	$\rightarrow I, \forall I$

1	$x, y, z \in A$	acp
2	$x \prec y$	acp
3	$y \prec z$	acp
4	$x \preceq y$	$\wedge E$
5	$y \preceq z$	$\wedge E$
6	$x \preceq z$	$trans$
7	$x \neq y$	$\wedge E$
8	$y \neq z$	$\wedge E$
9	$x = z$	acp
10	$y \preceq x$	$= E$
11	$x = y$	$antisym$
12	$\perp$	$\perp I$
13	$x \neq z$	$\neg I$
14	$x \prec z$	$\wedge I$
15	$x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I$
16	$\forall x, y, z \in A \quad x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I, \forall I$

2. if “ $\prec$ ” is a “(partial) Strict Order relation” in “ $\mathfrak{A}$ ” define “ $a \preceq b : a \prec b \vee a = b$ ” prove that “ $\preceq$ ” is “(partial) order relation”.

1	$x \in A$	acp
2	$x = x$	$= I$
3	$x \preceq x$	$\vee I$
4	$\forall x \in A \quad x \preceq x$	$\rightarrow I, \forall I$

1	$x, y \in A$	acp
2	$x \preceq y \wedge y \preceq x$	acp
3	$x \prec y \wedge y \prec x$	acp
4	$\perp$	$asym$
5	$x = y$	
6	$x \prec y \wedge y = x$	acp
7	$x = y$	
8	$x = y \wedge y \prec x$	acp
9	$x = y$	
10	$x = y \wedge y = x$	acp
11	$x = y$	
12	$x = y$	$\vee E$
13	$x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I$
14	$\forall x, y \in A \quad x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I, \forall I$

1	$x, y, z \in A$	<i>acp</i>
2	$x \preceq y \wedge y \preceq z$	<i>acp</i>
3	$x \prec y \wedge y \prec z$	<i>acp</i>
4	$x \prec z$	<i>trans</i>
5	$x \preceq z$	$\vee I$
6	$x \prec y \wedge y = z$	<i>acp</i>
7	$x \prec z$	$= E$
8	$x \preceq z$	$\vee I$
9	$x = y \wedge y \prec z$	<i>acp</i>
10	$x \prec z$	$= E$
11	$x \preceq z$	$\vee I$
12	$x = y \wedge y = z$	<i>acp</i>
13	$x = z$	$= E$
14	$x \preceq z$	$\vee I$
15	$x \preceq z$	$\vee E$
16	$x \preceq y \wedge y \prec z \rightarrow x \preceq z$	$\rightarrow I$
17	$\forall x, y, z \in A \quad x \preceq y \wedge y \prec z \rightarrow x \preceq z$	$\rightarrow I, \forall I$

definition 5.7.3. (Total-Linear) Order relation assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). we say “ $\preceq$ ” is a “(Total-Linear) Order relation” in “ $\mathfrak{A}$ ” if every two elements are comparable.

$$(\text{comparability}) : \forall x, y \in A \quad x \preceq y \vee y \preceq x$$

definition 5.7.4. (Total-Linear) Strict order relation assume a structure “ $\mathfrak{A} := (A, \prec)$ ” (“ $\prec$ ” is a Strict order relation). we say “ $\prec$ ” is a “(Total-Linear) Strict order relation” in “ $\mathfrak{A}$ ” if every two elements are comparable.

$$(\text{comparability}) : \forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$$

Theorem 5.7.2. .

1. if “ $\preceq$ ” is a “(Total-Linear) Order relation” in “ $\mathfrak{A}$ ” define “ $a \prec b : a \preceq b \wedge a \neq b$ ” prove that “ $\prec$ ” is “(Total-Linear) Strict order relation”.
Proof Is Left As An Exercise To The Reader. 😊

2. if “ $\prec$ ” is a “(Total-Linear) Strict Order relation” in “ $\mathfrak{A}$ ” define “ $a \preceq b : a \prec b \vee a = b$ ” prove that “ $\preceq$ ” is “(Total-Linear) order relation”.
Proof Is Left As An Exercise To The Reader. 😊

Lemma 5.7.1. assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). and $B, C \subseteq A$ prove

1. $\exists b \in B \quad \forall x \in B \quad b \preceq x / \therefore \exists! b \in B \quad \forall x \in B \quad b \preceq x$
Proof Is Left As An Exercise To The Reader. 😊. in this case we say “B” has least element or Minimum or “HaveLeast(B)”
2. $\exists c \in C \quad \forall x \in C \quad x \preceq c / \therefore \exists! c \in C \quad \forall x \in C \quad x \preceq c$
Proof Is Left As An Exercise To The Reader. 😊. in this case we say “C” has greatest element or has Maximum or “HaveGreatest(C)”

Theorem 5.7.3. Minimum and Maximum assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid \text{HaveLeast}(B)\}$ and $\mathcal{C} := \{C \subseteq A \mid \text{HaveGreatest}(C)\}$ we define like this

$$\min_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times V \mid \forall x \in B \quad b \preceq x\}$$

$$\max_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times V \mid \forall x \in C \quad x \preceq c\}$$

prove these two relations are two functions “ $\min_{\mathfrak{A}} : \mathcal{B} \rightarrow V$ ” and “ $\max_{\mathfrak{A}} : \mathcal{C} \rightarrow V$ ” 😊

definition 5.7.5. Minimal and Maximal assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). we define “Minimal and Maximal” like this

$$\text{Minimal}_{\mathfrak{A}}(b) : \forall x \in A \quad x \not\prec b$$

i.e. “b” is the minimal in “ $\mathfrak{A}$ ”.

$$\text{Maximal}_{\mathfrak{A}}(c) : \forall x \in A \quad c \not\prec x$$

i.e. “c” is the maximal in “ $\mathfrak{A}$ ”.

definition 5.7.6. Well ordering assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). we say “ $\mathfrak{A}$ ” is “Well ordered” when every nonempty subset of “A” has a “Minimum” i.e.

$$\text{WellOrdered}(\mathfrak{A}) : \forall X \subseteq A \quad X \neq \emptyset \rightarrow \text{HaveLeast}(X)$$

definition 5.7.7. Lower and Upper Bound assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). $B, C \subseteq A$. we define “Lower and Upper Bound” like this

$$\text{Lowerb}_{\mathfrak{A}}(b, B) : \forall x \in B \quad b \preceq x$$

i.e. “b” is the lowerbound of B in “ $\mathfrak{A}$ ”.

$$\text{Upperb}_{\mathfrak{A}}(c, C) : \forall x \in C \quad x \preceq c$$

i.e. “C” is the upperbound of B in “ $\mathfrak{A}$ ”.

if the stucture clear to us we don’t write stucture index “ $\min(_)$, $\text{Minimal}(_)$, $\text{Lowerb}(_, _)$ ”. and we don’t mension it.

Theorem 5.7.4. Infimum and Supremum assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid \text{HaveGreatest}\{b \in A \mid \text{Lowerb}_{\mathfrak{A}}(b, B)\}\}$ and $\mathcal{C} := \{C \subseteq A \mid \text{HaveLeast}\{c \in A \mid \text{Upperb}_{\mathfrak{A}}(c, C)\}\}$ we define “ $\text{inf}_{\mathfrak{A}} : \mathcal{B} \rightarrow V$ ” and “ $\text{sup}_{\mathfrak{A}} : \mathcal{C} \rightarrow V$ ” like this

$$\text{inf}_{\mathfrak{A}}(B) := \max_{\mathfrak{A}}\{b \in A \mid \text{Lowerb}_{\mathfrak{A}}(b, B)\}$$

$$\text{sup}_{\mathfrak{A}}(B) := \min_{\mathfrak{A}}\{c \in A \mid \text{Upperb}_{\mathfrak{A}}(c, C)\}$$

Theorem 5.7.5. assume a structure “ $\mathfrak{A} := (A, \preceq)$ ” (“ $\preceq$ ” is a Order relation). and $B \subseteq A$ prove

1. $\text{min}_{\mathfrak{A}}(B) = b / \therefore \text{inf}_{\mathfrak{A}}(B) = b$.

Proof Is Left As An Exercise ToThe Reader. 😊

2. $\text{inf}_{\mathfrak{A}}(B) = b, b \in B / \therefore \text{min}_{\mathfrak{A}}(B) = b$.

Proof Is Left As An Exercise ToThe Reader. 😊

this theorem holds for “ $\max, \text{sup}$ ” too.

This page intentionally left blank.

Chapter 6

The beginning of Abstract Algebra

we want to introduce some functions " $\circ : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ " to you. we use " $m \circ n$ " instead of " $\circ(m, n)$ ".

6.1 Group theory

definition 6.1.1. A structure $\mathfrak{G} := (G, \cdot)$

1. If $\forall x, y, z \in G \quad (xy)z = x(yz)$ (associative $\times$), $\mathfrak{G}$ is a **semigroup**.
2. If $\mathfrak{G}$ is a **semigroup** and $\forall x \in G \quad \exists y \in G \quad xy = yx = x$ (has identity $\times$), $\mathfrak{G}$ is a **monoid**.
you can prove for every **monoid** $\mathfrak{G}$ identity element is unique we denote it by " e " (in $\times$ with " 1 " and in $+$ with 0).
3. If $\mathfrak{G}$ is a **monoid** and $\forall x \in G \quad \exists y \in G \quad xy = yx = e$ (has inverse $\times$), $\mathfrak{G}$ is a **group**.
4. If $\mathfrak{G}$ is a **semigroup/monoid** and $\forall x, y, z \in N \quad xz = yz \rightarrow x = y$ (cancellative $\times$), $\mathfrak{G}$ is a **cancellative semigroup/monoid**. (you can prove every group is cancellative as an exercise. 😊)
5. If $\mathfrak{G}$ is a **semigroup/monoid/group** and $\forall x, y \in G \quad xy = yx$ (commutative(abelian) $\times$), $\mathfrak{G}$ is a **commutative(abelian) semigroup/monoid/group**.

6.2 Ring theory

definition 6.2.1. A structure $\mathfrak{R} := (R, +, \cdot)$

1. If $(R, +)$ is a abelian **monoid/group** and $\forall x, y, z \in R \quad (xy)z = x(yz)$ (associative $\times$),
 $\forall x, y, z \in R \quad \left((x(y+z) = xy + xz) \wedge ((x+y)z = xz + yz) \right)$ (distributive $\times$), $\mathfrak{R}$ is a **Semiring/Ring**.

2. If $\mathfrak{R}$ is a **Semiring/Ring** and $\forall x \in R \quad \exists y \in R \quad xy = yx = x$ (has identity $\times$), $\mathfrak{R}$ is a **Semiring/Ring with identity**. you can prove for every **Semiring/Ring with identity** $\mathfrak{R}$ identity element is unique we denote it by “1”.
3. If $\mathfrak{R}$ is a **Semiring/Ring** and $\forall x, y \in R \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (zero divisor free $\times$), $\mathfrak{R}$ is a **zero divisor free Semiring/Ring**. (you can prove every **division Semiring/Ring** is zero divisor free as an exercise. 😊)
4. If $\mathfrak{R}$ is a **Semiring/Ring with identity** and $\forall x \in R \quad \exists y \in R \quad xy = yx = 1$ (has inverse $\times$), $\mathfrak{R}$ is a **division Semiring/Ring** with.
5. If $\mathfrak{R}$ is a **Semiring/Ring** and $\forall x, y \in R \quad xy = yx$ (commutative(abelian) $\times$), $\mathfrak{R}$ is a **commutative(abelian) Semiring/Ring**.
6. If $\mathfrak{R}$ is a **zero divisor free Semiring/Ring** and **commutative(abelian) Semiring/Ring**. $\mathfrak{R}$ is a **Semidomain/Domain**.
7. If $\mathfrak{R}$ is a **division Semiring/Ring** and **commutative(abelian) Semiring/Ring**. $\mathfrak{R}$ is a **SemiField/Field**.

Theorem 6.2.1 (Universal Ring Completion of a Semiring). (Standard name: “Grothendieck Ring Completion Theorem for Commutative Semirings”) Let S be a (not necessarily commutative) semiring. There exists a ring R together with a semiring homomorphism

$$\eta : S \longrightarrow R$$

which is universal among semiring-to-ring homomorphisms (i.e. every homomorphism $f : S \rightarrow T$ into a ring T factors uniquely through η) if and only if the additive monoid $(S, +, 0)$ is a cancellative commutative monoid.

In this case, R is the Grothendieck group $G(S)$ of the additive monoid of S , with multiplication extending that of S by

$$[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)].$$

Proof. We divide the proof into two parts.

(1) Necessity of cancellativity and commutativity of $+$

Suppose $\eta : S \rightarrow R$ is an embedding of S as a subsemiring of a ring R . The additive structure of a ring is an abelian group. Thus the image $\eta(S)$ must be closed under an abelian addition. Therefore $(S, +)$ must already be a commutative monoid.

If $a + c = b + c$ in S , then in R we have

$$\eta(a) + \eta(c) = \eta(b) + \eta(c).$$

Cancelling $\eta(c)$ in the ring gives $\eta(a) = \eta(b)$, hence $a = b$ since η is injective. Thus $(S, +)$ must be cancellative.

This proves necessity.

(2) Construction of the universal ring when $(S, +)$ is cancellative and commutative

Assume now that $(S, +)$ is a cancellative commutative monoid. We form its Grothendieck group:

$$G(S) = (S \times S) / \sim,$$

where the relation is

$$(a, b) \sim (a', b') \iff a + b' = a' + b.$$

Equivalence relation

Reflexivity and symmetry are immediate. For transitivity, assume

$$a + b' = a' + b, \quad a' + b'' = a'' + b'.$$

Add b to the second equation and b'' to the first, and use commutativity and cancellativity to deduce

$$a + b'' = a'' + b.$$

Thus $\sim$ is an equivalence relation.

Well-defined addition

Define

$$[(a, b)] + [(c, d)] := [(a + c, b + d)].$$

If $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, then

$$a + b' = a' + b, \quad c + d' = c' + d.$$

Compute:

$$(a + c) + (b' + d') = (a + b') + (c + d') = (a' + b) + (c' + d) = (a' + c') + (b + d).$$

Thus $(a + c, b + d) \sim (a' + c', b' + d')$, so addition is well defined.

Well-defined multiplication

Define

$$[(a, b)] \cdot [(c, d)] := [(ac + bd, ad + bc)].$$

Assume $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, so

$$a + b' = a' + b, \quad c + d' = c' + d.$$

Expanding distributively:

$$(ac + bd) + (a'd' + b'c') = (a + b')(c + d') = (a' + b)(c' + d) = (a'c' + b'd') + (ad + bc).$$

Thus multiplication is well defined.

Ring structure

The additive structure is an abelian group by construction. Multiplication is associative and distributive because it is defined via bilinearity inherited from S . The multiplicative identity is $[(1, 0)]$.

Thus $G(S)$ is a ring.

(3) Universality

Given any semiring-to-ring homomorphism $f : S \rightarrow T$, define

$$\bar{f} : G(S) \rightarrow T, \quad \bar{f}([(a, b)]) := f(a) - f(b).$$

If $(a, b) \sim (a', b')$, then $a + b' = a' + b$, so applying f gives

$$f(a) + f(b') = f(a') + f(b),$$

hence $f(a) - f(b) = f(a') - f(b')$. Thus $\bar{f}$ is well defined. It is easily checked to be a ring homomorphism, and satisfies $\bar{f} \circ \eta = f$. Uniqueness follows since $[(a, b)] = [(a, 0)] - [(b, 0)]$.

This proves the universal property.

The Field of Fractions

The field of fractions is a fundamental concept in abstract algebra that allows us to extend a given integral domain to a field. It is the smallest field containing the integral domain.

Integral Domain

An integral domain is a commutative ring with a multiplicative identity ($1 \neq 0$) and no zero divisors. That is, if $a \cdot b = 0$, then either $a = 0$ or $b = 0$. The set of integers, $\mathbb{Z}$, is a classic example.

Field

A field is a commutative ring with a multiplicative identity where every non-zero element has a multiplicative inverse. For every $a \neq 0$, there exists b such that $a \cdot b = 1$. The set of rational numbers, $\mathbb{Q}$, is a field.

Construction of the Field of Fractions

Given an integral domain D , its field of fractions, denoted by $F(D)$, is constructed by formally creating fractions of elements from D . We consider pairs (a, b) where $a, b \in D$ and $b \neq 0$. An equivalence relation $\sim$ is defined on these pairs:

$$(a, b) \sim (c, d) \iff ad - bc = 0$$

The elements of $F(D)$ are the equivalence classes of these pairs, denoted by $\frac{a}{b}$.

Operations in $F(D)$: For $\frac{a}{b}, \frac{c}{d} \in F(D)$:

- Addition: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$
- Multiplication: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

The original integral domain D is embedded into $F(D)$ via the map $\phi : D \rightarrow F(D)$ defined by $\phi(a) = \frac{a}{1}$.

□

Theorem: Existence and Uniqueness of the Field of Fractions

Theorem 6.2.2. *Let D be an integral domain. Then there exists a field F and an injective ring homomorphism $\phi : D \rightarrow F$ such that for every field K and every ring homomorphism $\psi : D \rightarrow K$, there exists a unique field homomorphism $f : F \rightarrow K$ such that $\psi = f \circ \phi$.*

Proof. Let D be an integral domain.

Part 1: Existence

Consider the set $S = \{(a, b) \mid a, b \in D, b \neq 0\}$. Define an equivalence relation $\sim$ on S :

$$(a, b) \sim (c, d) \iff ad - bc = 0$$

The relation is reflexive, symmetric, and transitive because D is an integral domain. Let $F(D)$ be the set of equivalence classes of S under $\sim$. We denote the equivalence class of (a, b) by $\frac{a}{b}$.

Operations:

- Addition: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$
- Multiplication: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$

These operations are well-defined. For example, for addition, if $\frac{a}{b} = \frac{a'}{b'}$ and $\frac{c}{d} = \frac{c'}{d'}$, then $ab' = ba'$ and $cd' = dc'$. We must show $\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'}$, which means showing $(ad+bc)b'd' = (a'd'+b'c')bd$. This can be verified algebraically using the properties of D . Similarly for multiplication.

Field Axioms:

- $F(D)$ is closed under addition and multiplication.
- Addition and multiplication are associative, commutative, and distributive because these properties hold in D .
- The additive identity is $\frac{0}{1}$: $\frac{a}{b} + \frac{0}{1} = \frac{a \cdot 1 + b \cdot 0}{b \cdot 1} = \frac{a}{b}$.
- The multiplicative identity is $\frac{1}{1}$: $\frac{a}{b} \cdot \frac{1}{1} = \frac{a \cdot 1}{b \cdot 1} = \frac{a}{b}$.
- The additive inverse of $\frac{a}{b}$ is $\frac{-a}{b}$: $\frac{a}{b} + \frac{-a}{b} = \frac{0}{b} = \frac{0}{1}$.
- The multiplicative inverse of a non-zero element $\frac{a}{b}$ (where $a \neq 0$) is $\frac{b}{a}$: $\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ba} = \frac{1}{1}$.

Thus, $F(D)$ is a field.

Injective Ring Homomorphism $\phi : D \rightarrow F(D)$: Define $\phi(a) = \frac{a}{1}$.

- $\phi(a + b) = \frac{a+b}{1} = \frac{a}{1} + \frac{b}{1} = \phi(a) + \phi(b)$.
- $\phi(ab) = \frac{ab}{1} = \frac{a}{1} \cdot \frac{b}{1} = \phi(a) \cdot \phi(b)$.
- ϕ is injective: If $\phi(a) = \phi(b)$, then $\frac{a}{1} = \frac{b}{1}$, which implies $a \cdot 1 = b \cdot 1$, so $a = b$.

This establishes the existence of $F(D)$ and ϕ .

Part 2: Uniqueness

Suppose K is another field and $\psi : D \rightarrow K$ is an injective ring homomorphism satisfying the universal property. We need to define a unique field homomorphism $f : F(D) \rightarrow K$ such that $\psi = f \circ \phi$.

Define f : For any $\frac{a}{b} \in F(D)$, define

$$f\left(\frac{a}{b}\right) = \psi(a)(\psi(b))^{-1}$$

Since ψ is injective and $b \neq 0$, $\psi(b) \neq 0$, so $(\psi(b))^{-1}$ exists in K . This map f is well-defined, as shown in the detailed explanation (if $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$, and applying ψ gives $\psi(a)\psi(d) = \psi(b)\psi(c)$, leading to $\psi(a)(\psi(b))^{-1} = \psi(c)(\psi(d))^{-1}$).

f is a Field Homomorphism:

- $f\left(\frac{a}{b} + \frac{c}{d}\right) = f\left(\frac{ad+bc}{bd}\right) = \psi(ad+bc)(\psi(bd))^{-1} = (\psi(a)\psi(d) + \psi(b)\psi(c))(\psi(b)\psi(d))^{-1} = \psi(a)(\psi(b))^{-1} + \psi(c)(\psi(d))^{-1} = f\left(\frac{a}{b}\right) + f\left(\frac{c}{d}\right)$.
- $f\left(\frac{a}{b} \cdot \frac{c}{d}\right) = f\left(\frac{ac}{bd}\right) = \psi(ac)(\psi(bd))^{-1} = (\psi(a)\psi(c))(\psi(b)\psi(d))^{-1} = (\psi(a)(\psi(b))^{-1})(\psi(c)(\psi(d))^{-1}) = f\left(\frac{a}{b}\right)f\left(\frac{c}{d}\right)$.

Check Composition: For $a \in D$, $(f \circ \phi)(a) = f(\phi(a)) = f\left(\frac{a}{1}\right) = \psi(a)(\psi(1))^{-1} = \psi(a) \cdot 1_K^{-1} = \psi(a)$. So $\psi = f \circ \phi$.

Uniqueness of f : Suppose $g : F(D) \rightarrow K$ is another field homomorphism such that $\psi = g \circ \phi$. For any $\frac{a}{b} \in F(D)$, we have $\frac{a}{b} = \frac{a}{1} \cdot \left(\frac{b}{1}\right)^{-1}$ in $F(D)$. Then $g\left(\frac{a}{b}\right) = g\left(\frac{a}{1} \cdot \left(\frac{b}{1}\right)^{-1}\right) = g\left(\frac{a}{1}\right) \cdot g\left(\left(\frac{b}{1}\right)^{-1}\right) = g\left(\frac{a}{1}\right) \cdot (g\left(\frac{b}{1}\right))^{-1}$. Since $\psi = g \circ \phi$, we have $g\left(\frac{a}{1}\right) = \psi(a)$ and $g\left(\frac{b}{1}\right) = \psi(b)$. So, $g\left(\frac{a}{b}\right) = \psi(a)(\psi(b))^{-1}$, which is precisely $f\left(\frac{a}{b}\right)$. Thus, $f = g$, and f is unique. $\square$

Chapter 7

Natural numbers

7.1 Introduction to Natural numbers

definition 7.1.1. *Natural numbers* we define “0, 1, 2, 3, ...” like this.

$$\begin{aligned} 0 &:= \emptyset \\ 1 &:= \{0\} = \emptyset \cup \{\emptyset\} = \{\emptyset\} \\ 2 &:= \{0, 1\} = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\} \\ 3 &:= \{0, 1, 2\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \\ &\vdots \end{aligned}$$

definition 7.1.2. *Successor function* is a function “ $s : V \rightarrow V$ ” defined like this.

$$s(x) := x \cup \{x\}$$

you can see $\forall x \quad \text{Type}_0(x) \rightarrow \text{Type}_0(s(x))$

definition 7.1.3. we define a set “**Inductive**” like this.

$$\text{Ind}(X) : \emptyset \in X \wedge \forall y (y \in X \rightarrow s(y) \in X)$$

notice that “ $\text{Ind}(X)$ ” might be empty.

Axiom 7.1.1. *The Axiom of Infinity* we define a set “Inductive” like this.

$$\exists X \quad \text{Type}_0(X) \wedge \text{Ind}(X)$$

so “ $\text{Ind}(X)$ ” is a nonempty set. so we can define

definition 7.1.4. *Set of Natural numbers*($\mathbb{N}$)

$$\mathbb{N} := \bigcap Ind$$

Lemma 7.1.1. *prove.*

$$1. \mathbb{N} = \{x \mid \underbrace{\forall Y \left(\underbrace{(\emptyset \in Y \wedge \forall z(z \in Y \rightarrow s(z) \in Y))}_{Ind(Y)} \rightarrow x \in Y \right)}_{\phi(x)}\}$$

$$2. Ind(\mathbb{N})$$

Proof Is Left As An Exercise To The Reader. 😊

notice that now “ $\phi(x)$ ” is a proposition. so “ $\{x \mid \forall Y \left((\emptyset \in Y \wedge \forall z(z \in Y \rightarrow s(z) \in Y)) \rightarrow x \in Y \right)\}$ ” is a set.

7.2 Properties of Natural numbers

Theorem 7.2.1. Induction $P(0), \forall n \in \mathbb{N} \left(P(n) \rightarrow P(s(n)) \right) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

$$\begin{array}{ll} 1 & P(0) \\ 2 & \forall n \in \mathbb{N} \quad \left(P(n) \rightarrow P(s(n)) \right) \\ 3 & \left| \begin{array}{l} n \in \mathbb{N} \\ \forall n \left(P(n) \rightarrow P(s(n)) \right) \\ Ind(P) \\ n \in \bigcap Ind \\ \forall X (Ind(X) \rightarrow n \in X) \\ Ind(P) \rightarrow P(n) \\ P(n) \end{array} \right. \\ 10 & n \in \mathbb{N} \rightarrow P(n) \\ 11 & \forall n \in \mathbb{N} \quad P(n) \end{array}$$

definition 7.2.1. A set “ T ” is **Transitive** if $u \in v \in T$ implies $u \in T$.

$$Transitive(T) : \forall u, v \quad u \in v \in T \rightarrow u \in T$$

Lemma 7.2.1. .

1. $\forall n \in \mathbb{N} \quad \text{Transitive}(n)$

2. $\text{Transitive}(\mathbb{N})$

Proof Is Left As An Exercise To The Reader. 😊

definition 7.2.2. Order in Natural numbers($\mathbb{N}$)

$$< := \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x \in y\}$$

Theorem 7.2.2. Order of Natural numbers($\mathbb{N}$) *prove*

1. $\forall n \in \mathbb{N} \quad (n, <) \text{ is Well ordered.}$

2. $(\mathbb{N}, <) \text{ is lineary ordered.}$

3. $(\mathbb{N}, <) \text{ is Well ordered.}$

4. $(\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x < y) \quad (\text{doesn't have right endpoint})$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 7.2.3. Induction ,second vrsion $\forall k < n (P(k) \rightarrow P(n)) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

7.3 The Recurtion theorem

definition 7.3.1. Tuple (new definition)-Finite sequence is a function " $a : n \rightarrow V$ " s.t. $n \in \mathbb{N}$ we know we can show this by " $(a_i)_{i \in n}$ " we can also show it by " $(a_0, a_1, a_2, \dots, a_{n-1})$ "

definition 7.3.2. Sequence is a function " $a : \mathbb{N} \rightarrow V$ " we know we can show this by " $(a_i)_{i \in \mathbb{N}}$ " we can also show it by " $(a_i)_{i=0}^\infty$ "

now we can redefine "Cartesian product". we define finite Cartesian product " $\prod_{i \in n} A_i := \{(a_i)_{i \in n} \mid a_i \in A_i\}$ " and define infinite Cartesian product " $\prod_{i \in \mathbb{N}} A_i := \{(a_i)_{i \in \mathbb{N}} \mid a_i \in A_i\}$ " and define Cartesian product in general way " $\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i\}$ ". and you can easily see that from the definition of " B^A ", " A^n " is the set of all n -length finite sequences of $a \in A$ and " $A^\mathbb{N}$ " is the set of all infinite sequences of $a \in A$.

Theorem 7.3.1 (Recursion). *For any set A , any $a \in A$, and any function $g : A \times \mathbb{N} \rightarrow A$, there is a unique function $f : \mathbb{N} \rightarrow A$ such that:*

1. $f(0) = a.$

2. $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n).$

Proof. Step 1: Construction of F and f .

Define

$$F := \left\{ t \mid \exists m \in \mathbb{N} \left(t : s(m) \rightarrow A \wedge t(0) = a \wedge \forall k < m \left(t(s(k)) = g(t(k), k) \right) \right) \right\}.$$

Define

$$f := \bigcup F.$$

Claim 1: $F \neq \emptyset$.

We construct, by induction on $n \in \mathbb{N}$, a sequence of functions $t_n : s(n) \rightarrow A$ such that:

$$t_n(0) = a, \quad \forall k < n \quad t_n(s(k)) = g(t_n(k), k).$$

- **Base:** For $n = 0$, $s(0)$ is the set whose only element is 0. Define $t_0 : s(0) \rightarrow A$ by $t_0(0) = a$. The condition on $k < 0$ is vacuous, so t_0 satisfies the required property, hence $t_0 \in F$.
- **Step:** Suppose $t_n : s(n) \rightarrow A$ is defined with the desired properties. Define $t_{n+1} : s(n+1) \rightarrow A$ by:

$$t_{n+1}(k) := \begin{cases} t_n(k) & \text{if } k \in s(n), \\ g(t_n(n), n) & \text{if } k = s(n). \end{cases}$$

Then $t_{n+1}(0) = t_n(0) = a$, and for all $k < n+1$:

$$t_{n+1}(s(k)) = \begin{cases} t_n(s(k)) = g(t_n(k), k) & \text{if } k < n, \\ g(t_n(n), n) & \text{if } k = n, \end{cases}$$

so t_{n+1} also satisfies the recursion condition. Hence $t_{n+1} \in F$.

Thus, for every $n \in \mathbb{N}$, $t_n \in F$, so $F \neq \emptyset$ and $f = \bigcup F$ is well-defined.

Claim 2: f is a function.

We must show $\text{Func}(f)$, i.e., for all n , if $(n, x) \in f$ and $(n, y) \in f$, then $x = y$.

Let $(n, x) \in f$ and $(n, y) \in f$. Then there exist $t, u \in F$ such that $(n, x) \in t$ and $(n, y) \in u$. Let $\text{Dom}(t) = s(m)$ and $\text{Dom}(u) = s(\ell)$. Without loss of generality, assume $m \leq \ell$.

We show $t \subseteq u$ by induction on $k < m$.

- **Base:** $k = 0$. Then $t(0) = a$ and $u(0) = a$ by definition of F , so $t(0) = u(0)$.
- **Step:** Assume $t(k) = u(k)$ for some $k < m$. Since $t, u \in F$, both satisfy the recursion condition:

$$t(s(k)) = g(t(k), k), \quad u(s(k)) = g(u(k), k).$$

By the induction hypothesis, $t(k) = u(k)$, hence $t(s(k)) = u(s(k))$.

Thus t and u agree on all of $s(m)$, so in particular $t(n) = u(n)$ whenever $n \in s(m)$. Hence $x = t(n) = u(n) = y$. Therefore f is a function:

$$\text{Func}(f).$$

Claim 3: $\text{Dom}(f) = \mathbb{N}$ and $\text{Range}(f) \subseteq A$.

First, for each $n \in \mathbb{N}$, the function $t_n : s(n) \rightarrow A$ constructed above belongs to F , and $n \in \text{Dom}(t_n)$. Thus $(n, t_n(n)) \in f$, so $n \in \text{Dom}(f)$. Hence

$$\mathbb{N} \subseteq \text{Dom}(f).$$

Conversely, if $n \in \text{Dom}(f)$, then there exists $t \in F$ such that $n \in \text{Dom}(t)$. But $\text{Dom}(t) = s(m)$ for some $m \in \mathbb{N}$, so $n \in s(m) \subseteq \mathbb{N}$. Thus $\text{Dom}(f) \subseteq \mathbb{N}$, and therefore

$$\text{Dom}(f) = \mathbb{N}.$$

For the range, if $(n, x) \in f$, then $(n, x) \in t$ for some $t \in F$, and each such t has codomain A . Hence $x \in A$, so

$$\text{Range}(f) \subseteq A.$$

Claim 4: f satisfies the recursion equations.

(1) $f(0) = a$.

For every $t \in F$, we have $t(0) = a$ by definition of F . Since $0 \in \text{Dom}(t)$ for all $t \in F$, and all these values agree, the union $f = \bigcup F$ also satisfies

$$f(0) = a.$$

(2) $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$.

Fix $n \in \mathbb{N}$. Consider the function $t_{n+1} \in F$ with domain $s(n+1)$. Then $n, s(n) \in \text{Dom}(t_{n+1})$, and by construction:

$$t_{n+1}(s(n)) = g(t_{n+1}(n), n).$$

Since $t_{n+1} \subseteq f$, we have

$$f(n) = t_{n+1}(n), \quad f(s(n)) = t_{n+1}(s(n)).$$

Therefore

$$f(s(n)) = g(f(n), n).$$

Thus f satisfies both required conditions.

Uniqueness.

Suppose $h : \mathbb{N} \rightarrow A$ is another function such that:

1. $h(0) = a$,
2. $\forall n \in \mathbb{N} \quad h(s(n)) = g(h(n), n)$.

We show $h = f$ by induction on n .

- **Base:** $n = 0$. Then $h(0) = a = f(0)$.
- **Step:** Assume $h(n) = f(n)$ for some $n \in \mathbb{N}$. Then

$$h(s(n)) = g(h(n), n) = g(f(n), n) = f(s(n)).$$

Thus, by induction, $h(n) = f(n)$ for all $n \in \mathbb{N}$, so $h = f$.

Therefore, there exists a unique function $f : \mathbb{N} \rightarrow A$ such that $f(0) = a$ and $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$. $\square$

Theorem 7.3.2. (Parametric version) recursion for any set “ A, P ” and “ $a : P \rightarrow A$ ” and any function “ $g : P \times A \times \mathbb{N} \rightarrow A$ ” there is a unique function “ $f : P \times \mathbb{N} \rightarrow A$ ” s.t.

1. $f(p, 0) = a(p)$
2. $\forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(n), n)$

Claim 1 : Existing of “ f ”

Claim 2 : “ f ” is a function

Claim 3 : “ $\text{Dom}(f) = \mathbb{N}$ ” and “ $\text{Range}(f) \subseteq A$ ”

Claim 4 : “ f ” satisfies conditions (1) and (2)

7.4 Arithmetic of Natural numbers

Theorem 7.4.1. Summation there is a unique function “ $+$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ ” s.t.

1. $\forall m \in \mathbb{N} \quad m + 0 = m$
2. $\forall m, n \in \mathbb{N} \quad m + s(n) = s(m + n)$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 7.4.2. Multiplication there is a unique function “ $\times$: $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ ” s.t.

1. $\forall m \in \mathbb{N} \quad m \times 0 = 0$
2. $\forall m, n \in \mathbb{N} \quad m \times s(n) = m \times n + m$

Proof Is Left As An Exercise To The Reader. 😊

we denote also “Multiplication” with “ $m \cdot n$ ” and “ mn ” too.

7.5 Peano arithmetic

assume a structure “ $\mathfrak{N} := (N, +, \cdot, s, 0)$ ” (here “ $+$, “ $\cdot$ ”, “ s ”, “ 0 ” are arbitrary functions and constants and doesn’t have meaning we defined before). with these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \rightarrow x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x, y \in N \quad x + s(y) = s(x + y)$
5. $\forall x \in N \quad x \cdot 0 = 0$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\phi(0) \wedge \forall y \in N \quad (\phi(y) \rightarrow \phi(s(y))) \rightarrow \forall y \in N \quad \phi(y)$ (this is true for arbitrary “ ϕ ”)

we call “ $\mathfrak{N}$ ” a model of “Peano arithmetic”.

Lemma 7.5.1. *prove for “ $(N, +)$ ”*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ (*assosiative +*)
2. $\forall x \in N \quad x + 0 = 0 + x = x$ (*identity +*)
3. $\forall x, y, z \in N \quad x + z = y + z \rightarrow x = y$ (*cancellative +*)
4. $\forall x, y \in N \quad x + y = y + x$ (*commutative(abelian) +*)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 7.5.2. *prove for “ $(N, +, \cdot)$ ”*

1. $\forall x, y, z \in N \quad (xy)z = x(yz)$ (*assosiative $\times$*)
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ (*identity $\times$*)
3. $\forall x, y \in N \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (*zero divisor free $\times$*)
4. $\forall x, y \in N \quad mn = nm$ (*commutative(abelian) $\times$*)
5. $\forall x, y, z \in N \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (*distributive $\times$*)

Proof Is Left As An Exercise To The Reader. 😊

Now define “ $x < y : \exists z \neq 0 \quad x + z = y$ ”. You can prove for natural numbers

$$\forall m, n \in \mathbb{N} \quad m < n \leftrightarrow \exists k \neq 0 \quad m + k = n$$

Lemma 7.5.3. *prove for “ $(N, +, \times, <)$ ”*

1. $\forall x, y, z \in N \quad x < y \rightarrow x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

as you can see “ $(\mathbb{N}, +, \times, s, 0)$ ” is a model of peano arithmetic. so properties are true for them.

This page intentionally left blank.

Chapter 8

From Natural numbers to Complex numbers

8.1 Integers“ $\mathbb{Z}$ ”

we learned about “ $\mathbb{N}$ ” before and we constructed structure “ $(\mathbb{N}, +, \cdot, <)$ ”. if you assume “ $\mathbb{N} \times \mathbb{N}$ ” we can define an equivalence relation “ $(n_1, m_1) \sim (n_2, m_2) : n_1 + m_2 = n_2 + m_1$ ”(check it’s an equivalence relation).

definition 8.1.1. *Integers“ $\mathbb{Z}$ ” is a set*

$$\mathbb{Z} := \mathbb{N} \times \mathbb{N} / \sim$$

we denote it’s members with “ $z_0, z_1, \dots$ ”

in this system “ $0 := [(0, 0)]$ ” and “ $\forall n \in \mathbb{N} \quad n := [(n, 0)]$ ” and “ $\forall n \in \mathbb{N} \quad -n := [(0, n)]$ ”. in general “ $\forall m, n \in \mathbb{N} \quad m - n := [(m, n)]$ ”. so we can say “ $\mathbb{N} \subseteq \mathbb{Z}$ ”.

Theorem 8.1.1. *Summation* there is a unique function “ $+_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ ” s.t.

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] +_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 + m_2, n_1 + n_2)]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.1.2. *Multiplication* there is a unique function “ $\times_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ ” s.t.

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] \times_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 m_2 + n_1 n_2, m_1 n_2 + m_2 n_1)]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 8.1.2. *Order in $\mathbb{Z}$*

$$[(m_1, n_1)] <_{\mathbb{Z}} [(m_2, n_2)] : m_1 + n_2 < m_2 + n_1$$

this lemma let us use “<” instead of “ $<_{\mathbb{Z}}$ ”, “+” instead of “ $+_{\mathbb{Z}}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{Z}}$ ”.

Lemma 8.1.1. *prove.*

1. $\therefore \forall m, n \in \mathbb{N} \quad m <_{\mathbb{Z}} n \leftrightarrow m < n$
2. $\therefore \forall m, n \in \mathbb{N} \quad m +_{\mathbb{Z}} n = m + n$
3. $\therefore \forall m, n \in \mathbb{N} \quad m \times_{\mathbb{Z}} n = m \times n$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $z \cdot y$ ” and “ zy ” too.

Lemma 8.1.2. *prove for “ $(\mathbb{Z}, +)$ ”*

1. $\forall x, y, z \in \mathbb{Z} \quad (x + y) + z = x + (y + z)$ (*associative +*)
2. $\forall z \in \mathbb{Z} \quad z + 0 = 0 + z = z$ (*identity +*)
3. $\forall z \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad z + y = y + z = 0$ (*inverse +*)
4. $\forall x, y \in \mathbb{Z} \quad x + y = y + x$ (*commutative(abelian) +*)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.1.3. *prove for “ $(\mathbb{Z}, +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{Z} \quad (xy)z = x(yz)$ (*associative $\times$*)
2. $\forall z \in \mathbb{Z} \quad z \cdot 1 = 1 \cdot z = z$ (*identity $\times$*)
3. $\forall x, y \in \mathbb{Z} \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (*zero divisor free $\times$*)
4. $\forall x, y \in \mathbb{Z} \quad xy = yx$ (*commutative(abelian) $\times$*)
5. $\forall x, y, z \in \mathbb{Z} \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (*distributive $\times$*)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.1.3. Order of Integers($\mathbb{Z}$) *prove for “ $(\mathbb{Z}, +, \times, <_{\mathbb{Z}})$ ”*

1. *is lineary ordered.*
2. $(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad x < y) \wedge (\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad y < x)$ (*doesn't have endpoints*)
3. $\forall x, y, z \in \mathbb{Z} \quad x < y \rightarrow x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z} \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

8.2 Rational numbers “ $\mathbb{Q}$ ”

we learned about “ $\mathbb{Z}$ ” before and we constructed structure “ $(\mathbb{Z}, +, \cdot, <)$ ”. if you assume “ $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ ” we can define an equivalence relation “ $(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$ ”(check it's an equivalence relation).

definition 8.2.1. *Rational numbers* “ $\mathbb{Q}$ ” is a set

$$\mathbb{Q} := \mathbb{Z} \times (\mathbb{Z} - \{0\}) / \sim$$

we denote it's members with “ $q_0, q_1, \dots$ ”

in this system “ $0 := [(0, 1)]$ ” and “ $1 := [(1, 1)]$ ” and “ $\forall x \in \mathbb{Z} \quad x := [(x, 1)]$ ” and “ $\forall x \in \mathbb{Z} \quad \frac{1}{x} := [(1, x)]$ ”. in general “ $\forall x, y \in \mathbb{Z} \quad \frac{x}{y} := [(x, y)]$ ”. so we can say “ $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ ”.

Theorem 8.2.1. Summation there is a unique function “ $+_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ ” s.t.

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x, z)] +_{\mathbb{Q}} [(y, u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.2.2. Multiplication there is a unique function “ $\times_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ ” s.t.

$$\forall x, y, z, u \in \mathbb{Z} \quad [(x, z)] \times_{\mathbb{Q}} [(y, u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 8.2.2. Order in $\mathbb{Q}$.

if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x, z)] <_{\mathbb{Q}} [(y, u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x, z)] <_{\mathbb{Q}} [(y, u)] : x \cdot u > y \cdot z$$

this lemma let us use “ $<$ ” instead of “ $<_{\mathbb{Q}}$ ”, “ $+$ ” instead of “ $+_{\mathbb{Q}}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{Q}}$ ”.

Lemma 8.2.1. *prove.*

1. $\therefore \forall z, y \in \mathbb{Z} \quad z <_{\mathbb{Q}} y \leftrightarrow z < y$
2. $\therefore \forall z, y \in \mathbb{Z} \quad z +_{\mathbb{Q}} y = z + y$
3. $\therefore \forall z, y \in \mathbb{Z} \quad z \times_{\mathbb{Q}} y = z \times y$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $x \cdot y$ ” and “ xy ” too.

Lemma 8.2.2. *prove for “ $(\mathbb{Q}, +)$ ”*

1. $\forall x, y, z \in \mathbb{Q} \quad (x + y) + z = x + (y + z)$ (assosiative +)
2. $\forall x \in \mathbb{Q} \quad x + 0 = 0 + x = x$ (identity +)
3. $\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x + y = y + x = 0$ (inverse +)
4. $\forall x, y \in \mathbb{Q} \quad x + y = y + x$ (commutative(abelian) +)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.2.3. *prove for “ $(\mathbb{Q}, +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{Q} \quad (xy)z = x(yz)$ (associative $\times$)
2. $\forall x \in \mathbb{Q} \quad x \cdot 1 = 1 \cdot x = x$ (identity $\times$)
3. $\forall x \in \mathbb{Q} - \{0\} \quad \exists y \in \mathbb{Q} \quad xy = yx = 1$ (inverse $\times$)
4. $\forall x, y \in \mathbb{Q} \quad xy = yx$ (commutative(abelian) $\times$)
5. $\forall x, y, z \in \mathbb{Q} \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.2.3. Order of Rational numbers($\mathbb{Q}$) *prove for “ $(\mathbb{Q}, <_{\mathbb{Q}})$ ”*

1. *is lineary ordered.*
2. $(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x < y) \wedge (\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad y < x)$ (doesn't have endpoints)
3. $\forall x, y \in \mathbb{Q} \quad \exists z \in \mathbb{Q} \quad x \neq y \rightarrow x < z < y$ (dense)
4. $\forall x, y, z \in \mathbb{Q} \quad x < y \rightarrow x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q} \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

8.3 Real numbers “ $\mathbb{R}$ ”

before we we get to lesson define function “ $|_| : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ ” like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means “ $|_| := \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid (0 \leq x \rightarrow y = x) \wedge (x < 0 \rightarrow y = -x)\}$ ”. learn this notation because we want to use later.

we learned about “ $\mathbb{Q}$ ” before and we constructed structure “ $(\mathbb{Q}, +, \cdot, <)$ ”. now assume $\mathbb{Q}^{\mathbb{N}}$ (we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}$). we don't want it all we just want a subset.

definition 8.3.1. Cauchy sequence *is a sequence like “ $(a_i)_{i \in \mathbb{N}} \in \mathbb{Q}^{\mathbb{N}}$ ” s.t. elements of it get infinitely close to each other i.e.*

$$Cauchy(a_i)_{i \in \mathbb{N}} : \forall \epsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \epsilon$$

we usually give “ N ” natural numbers but it's not necessary.

we define an equivalence relation “ $(a_i)_{i \in \mathbb{N}} \sim (b_i)_{i \in \mathbb{N}} : \forall \epsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \epsilon$ ”(check it's an equivalence relation).

definition 8.3.2. *Real numbers “ $\mathbb{R}$ ” is a set*

$$\mathbb{R} := \text{Cauchy} / \sim$$

we denote it's members with “ $r_0, r_1, \dots$ ”

in this system “ $0 := [(0)_{i \in \mathbb{N}}]$ ” and “ $1 := [(1)_{i \in \mathbb{N}}]$ ” and “ $\forall q \in \mathbb{Q} \quad q := [(q)_{i \in \mathbb{N}}]$ ”. so we can say “ $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ ”.

Theorem 8.3.1. Summation *there is a unique function “ $+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ ” s.t.*

$$[(a_i)_{i \in \mathbb{N}}] +_{\mathbb{R}} [(b_i)_{i \in \mathbb{N}}] = [(a_i + b_i)_{i \in \mathbb{N}}]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.3.2. Multiplication *there is a unique function “ $\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ ” s.t.*

$$[(a_i)_{i \in \mathbb{N}}] \times_{\mathbb{R}} [(b_i)_{i \in \mathbb{N}}] = [(a_i b_i)_{i \in \mathbb{N}}]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 8.3.3. Order in $\mathbb{R}$.

$$[(a_i)_{i \in \mathbb{N}}] <_{\mathbb{R}} [(b_i)_{i \in \mathbb{N}}] : \exists \epsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \epsilon$$

this lemma let us use “ $<$ ” instead of “ $<_{\mathbb{R}}$ ”, “ $+$ ” instead of “ $+_{\mathbb{R}}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{R}}$ ”.

Lemma 8.3.1. *prove.*

1. $\therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $\therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $\therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $x \cdot y$ ” and “ xy ” too.

Lemma 8.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$ (*assosiative +*)
2. $\forall x \in \mathbb{R} \quad x + 0 = 0 + x = x$ (*identity +*)
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x + y = y + x = 0$ (*inverse +*)
4. $\forall x, y \in \mathbb{R} \quad x + y = y + x$ (*commutative(abelian) +*)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ (*assosiative $\times$*)
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ (*identity $\times$*)

3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ (inverse $\times$)
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ (commutative(abelian) $\times$)
5. $\forall x, y, z \in \mathbb{R} \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.3.3. Linear order of Real numbers($\mathbb{R}$) prove “ $(\mathbb{R}, <_{\mathbb{R}})$ ”

1. is linearly ordered.
2. $(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y) \wedge (\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x)$ (doesn't have endpoints)
3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \rightarrow x < z < y$ (dense)
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \emptyset \wedge \exists y \in \mathbb{R} \quad \text{Uppreb}(X, y)) \rightarrow (\exists y \in \mathbb{R} \quad \text{sup}(X) = y)$ (complete)
5. $\forall x, y, z \in \mathbb{R} \quad x < y \rightarrow x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

8.4 Complex numbers“ $\mathbb{C}$ ”

we learned about “ $\mathbb{R}$ ” before and we constructed structure “ $(\mathbb{R}, +, \cdot, <)$ ”.

definition 8.4.1. Complex numbers“ $\mathbb{C}$ ” is a set

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}$$

we denote it's members with “ $z_0, x, \dots$ ”

in this system “ $0 := (0, 0)$ ” and “ $\forall n \in \mathbb{R} \quad r := (r, 0)$ ” . in general “ $\forall x, y \in \mathbb{R} \quad x + iy := (x, y)$ ”.
so we can say “ $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$ ”.

Theorem 8.4.1. Summation there is a unique function “ $+_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ ” s.t.

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) +_{\mathbb{C}} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.4.2. Multiplication there is a unique function “ $\times_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ ” s.t.

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) \times_{\mathbb{C}} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader. 😊

this lemma let us use “ $+$ ” instead of “ $+_{\mathbb{C}}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{C}}$ ”.

Lemma 8.4.1. prove.

1. / $\therefore \forall x, y \in \mathbb{R} \quad x +_{\mathbb{C}} y = x + y$

$$2. \quad \therefore \forall x, y \in \mathbb{R} \quad x \times_{\mathbb{C}} y = x \times y$$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $z \cdot y$ ” and “ zy ” too.

Lemma 8.4.2. *prove for “ $(\mathbb{C}, +)$ ”*

1. $\forall x, y, z \in \mathbb{C} \quad (x + y) + z = x + (y + z)$ (associative +)
2. $\forall z \in \mathbb{C} \quad z + 0 = 0 + z = z$ (identity +)
3. $\forall z \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad z + y = y + z = 0$ (inverse +)
4. $\forall x, y \in \mathbb{C} \quad x + y = y + x$ (commutative(abelian) +)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.4.3. *prove for “ $(\mathbb{C}, +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{C} \quad (xy)z = x(yz)$ (associative $\times$)
2. $\forall z \in \mathbb{C} \quad z \cdot 1 = 1 \cdot z = z$ (identity $\times$)
3. $\forall x, y \in \mathbb{C} \quad xy = yx$ (commutative(abelian) $\times$)
4. $\forall x, y, z \in \mathbb{C} \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.4.3. Exponents *there is a unique function “ $(_)^{(_)} : \mathbb{C} \times \mathbb{N} \rightarrow \mathbb{C}$ ” s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall n \in \mathbb{N} \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.4.4. *prove for “ $(\mathbb{C}, +, \cdot, (_)^{(_)})$ ”*

1. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.4.4. Exponents *there is a unique function “ $(_)^{(_)} : \mathbb{C} \times \mathbb{Z} \rightarrow \mathbb{C}$ ” s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall m, n \in \mathbb{N} \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.4.5. *prove for “ $(\mathbb{C}, +, \cdot)$ ”*

1. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.4.5. Big Sum if “ $a : \mathbb{N} \rightarrow \mathbb{C}$ ” there is a unique function “ $\sum_{i=0}^{(\cdot)}(\cdot) : \mathbb{C} \times \mathbb{N} \rightarrow \mathbb{C}$ ” s.t.

1. $\forall a \quad \sum_{i=0}^0(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^n a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.4.6. prove for “ $(\mathbb{C}, +, \cdot)$ ”

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \sum_{i=0}^n a_n + \sum_{i=0}^n b_n = \sum_{i=0}^n (a_n + b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad k \sum_{i=0}^n a_n = \sum_{i=0}^n k a_n$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 8.4.6. Big Product if “ $a : \mathbb{N} \rightarrow \mathbb{C}$ ” there is a unique function “ $\prod_{i=0}^{(\cdot)}(\cdot) : \mathbb{C} \times \mathbb{N} \rightarrow \mathbb{C}$ ” s.t.

1. $\forall a \quad \prod_{i=0}^0(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^n a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 8.4.7. prove for “ $(\mathbb{C}, +, \cdot)$ ”

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \prod_{i=0}^n a_n \prod_{i=0}^n b_n = \prod_{i=0}^n (a_n b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad (\prod_{i=0}^n a_n)^k = \prod_{i=0}^n (a_n)^k$

Proof Is Left As An Exercise To The Reader. 😊

Chapter 9

Finite and Infinite sets

9.1 Equipotent and Order in sets

we talked about “Isomorphism” before now assume two Structures $\mathfrak{A} := (A)$ and $\mathfrak{B} := (B)$. because these Structures have only a set “Isomorphism” between them only means there is a bijection function $f : A \xrightarrow[\text{surj}]{\text{inj}} B$.

definition 9.1.1. *Equipotent* in situation above we say “A” and “B” are “Equipotent”. we denote it with “ $A \cong B$ ” i.e.

$$A \cong B : \exists f(f : A \xrightarrow[\text{surj}]{\text{inj}} B)$$

Lemma 9.1.1. “ $\cong$ ” is a equivalence relation in “ $(V, \cong)$ ”

Proof Is Left As An Exercise To The Reader. 😊

so “ $V / \cong$ ” is a partision for “V” and every partition equals to a $[A]_{\cong}$ we denote it by $[A]$.

definition 9.1.2. *Equipotent* if there is a function $f : A \xrightarrow{\text{inj}} B$. in situation above we say “A” is less than or equipotent “B”. we denote it with “ $A \preceq B$ ” i.e.

$$A \preceq B : \exists f(f : A \xrightarrow{\text{inj}} B)$$

similarly we define “ $[A] \preceq [B] : \forall X \in [A], Y \in [B] \quad X \preceq Y$ ”

Lemma 9.1.2. $A \preceq B \therefore / \therefore [A] \preceq [B]$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 9.1.1. Schroeder-Bernstein $A \preceq B, B \preceq A / \therefore A \cong B$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 9.1.3. “ $\preceq$ ” is a order relation in “ $(V / \cong, \preceq)$ ”

Proof Is Left As An Exercise To The Reader. 😊

9.2 Finite sets

definition 9.2.1. if there is a “ $n \in \mathbb{N}$ ” s.t. $A \cong n$ we call it “**Finite**”(Finite(A)). if a set is not finite we call it “**Infinite**”(Infinite(A)).

Lemma 9.2.1. $\therefore \forall n \in \mathbb{N} \quad X \subseteq n \rightarrow X \not\cong n$

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.2.1. .

$$1. \forall m, n \in \mathbb{N} \quad m \neq n \rightarrow m \not\cong n$$

$$2. \forall m, n \in \mathbb{N} \quad A \cong m \wedge A \cong n \rightarrow m = n$$

$$3. \text{Infinite}(\mathbb{N})$$

$$4. \forall m, n \in \mathbb{N} \quad m \prec n \leftrightarrow m < n$$

Proof Is Left As An Exercise To The Reader. 😊

corollary (4) let us use “ $<$ ” instead of “ $\prec$ ”

Lemma 9.2.2. Finite(X), $Y \subseteq X \therefore$ Finite(Y)

Proof Is Left As An Exercise To The Reader. 😊

9.3 Countable sets

definition 9.3.1. set “ A ” is “**Countable**” if “ $A \cong \mathbb{N}$ ”. set “ A ” is “**At most countable**” if “ $A \preceq \mathbb{N}$ ”

Lemma 9.3.1. Countable(X), Infinite(Y), $Y \subseteq X \therefore$ Countable(Y)

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.1. a set is at most countable iff it’s finite or countable.

Lemma 9.3.2. range of infinite sequence is atmost countable.

Proof Is Left As An Exercise To The Reader. 😊

Lemma 9.3.3. union of two countable set is countable.

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.2. finite union of countable sets is countable.

Lemma 9.3.4. if “ A ” and “ B ” are countable sets then “ $A \times B$ ” is countable.

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.3. finite cartesian product of countable sets is countable. and “ A^n ” is countable for any “ A ” countable and “ $n > 0$ ”.

Lemma 9.3.5. if “ $(A_i)_{i \in \mathbb{N}}$ ” is family of countable sets with a family of sequences “ $(a_i)_{i \in \mathbb{N}}$ ” s.t.

“ $\forall n \in \mathbb{N} \quad a_n : \mathbb{N} \xrightarrow[\text{surj}]{\text{inj}} A_n$ ” then “ $\bigcup_{i \in \mathbb{N}} A_i$ ” is countable.

Proof Is Left As An Exercise To The Reader. 😊

Lemma 9.3.6. *if “ A ” is countable set then “ $\bigcup_{n \in \mathbb{N}} A^n$ ” is countable.*

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.4. *set of all finite subsets of countable set is countable.*

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.5. $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof Is Left As An Exercise To The Reader. 😊

Corollary 9.3.6. *a partition of countable set is countable.*

Proof Is Left As An Exercise To The Reader. 😊

9.4 Uncountable sets

Lemma 9.4.1. $\therefore \mathbb{N} \prec 2^{\mathbb{N}}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 9.4.2. $\therefore 2^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N})$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 9.4.3. $\therefore 2^{\mathbb{N}} \cong \mathbb{R}$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 9.4.1. Cantor theorem $\therefore \forall X \text{ Type}_0(X) \rightarrow X \prec 2^X$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 9.4.2. $\therefore \forall X \text{ Type}_0(X) \rightarrow 2^X \cong \mathcal{P}(X)$

Proof Is Left As An Exercise To The Reader. 😊

This page intentionally left blank.

Chapter 10

Ordinals

10.1 Well ordered sets

definition 10.1.1. if “ $(W, <)$ ” is well ordered and $a \in W$ “**Initial segment**” is

$$W[a] := \{x \in W \mid x < a\}$$

before get to next lemma Let’s have a definition. assume “ $(W_1, <_1)$ ” and “ $(W_2, <_2)$ ” are two ordered structures function “ f ” is increasing when “ $f : W_1 \rightarrow W_2$ ” is a (weak) homomorphism.

Lemma 10.1.1. if “ $(W, <)$ ” is well ordered and $f : W \rightarrow W$ is increasing then $\forall x \in W \quad x \leq f(x)$. Proof Is Left As An Exercise To The Reader. 😊

Corollary 10.1.1. .

1. no well ordered structure is isomorph to it’s initial segment of itself.
2. each well orderd structure has only one automorphism, the identity.
3. if “ $(W_1, <_1)$ ” and “ $(W_2, <_2)$ ” are two isomorphic well ordered structures then the isomorphism is unique.

Proof Is Left As An Exercise To The Reader. 😊

Theorem 10.1.1. if “ $(W_1, <_1)$ ” and “ $(W_2, <_2)$ ” are two well ordered structures then exactly one of the following holds.

1. $(W_1, <_1) \cong (W_2, <_2)$
2. $\exists a \in W_1 \quad (W_1[a], <_1) \cong (W_2, <_2)$
3. $\exists a \in W_2 \quad (W_1, <_1) \cong (W_2[a], <_2)$

Proof Is Left As An Exercise To The Reader. 😊

10.2 Ordinal numbers

definition 10.2.1. *Order in sets*

$$<_A := \{(x, y) \in A \times A \mid x \in y\}$$

definition 10.2.2. *A set “ α ” is a **Ordinal** if*

1. α is transitive.
2. $(\alpha, <_\alpha)$ is well ordered.

we denote it by $Ord(\mathbb{N})$ so we have the set of ordinals(Ord).

Lemma 10.2.1. *Every natural number is an ordinal.*

Proof Is Left As An Exercise To The Reader. 😊

In ordinal numbers we show “ $\mathbb{N}$ ” with “ ω ”. so it’s just a renaming.

Lemma 10.2.2. *if α is an ordinal then $s(\alpha)$ is an ordinal too.*

Proof Is Left As An Exercise To The Reader. 😊

Lemma 10.2.3. *prove*

1. $Transitive(Ord)$.
2. $\forall \alpha, \beta \in Ord \quad \alpha \subset \beta \rightarrow \alpha \in \beta$.

Proof Is Left As An Exercise To The Reader. 😊

Theorem 10.2.1. *Order of Ordinals prove*

1. $(Ord, <)$ is lineary ordered.
2. $(Ord, <)$ is Well ordered.

Proof Is Left As An Exercise To The Reader. 😊

Theorem 10.2.2. *The natural numbers are exactly finite ordinals.*

Proof Is Left As An Exercise To The Reader. 😊

10.3 Transfinite Induction

10.4 Transfinite Recursion

Theorem 10.4.1. *Classic Summation there is a unique function “ $+_c : Ord \times Ord \rightarrow Ord$ ” s.t.*

1. $\forall \alpha \in Ord \quad \alpha +_c 0 = \alpha$
2. $\forall \alpha, \beta \in Ord \quad \alpha +_c s(\beta) = s(\alpha +_c \beta)$
3. $\forall \alpha, \beta \in Ord \quad limit(\beta) \rightarrow \alpha +_c \beta = \bigcup_{\beta' \in \beta} (\alpha +_c \beta')$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 10.4.1. $\forall \alpha, \beta \in \text{Ord} (\alpha \leq \beta \rightarrow \exists \gamma \in \text{Ord} \quad \alpha +_c \gamma = \beta)$

Proof Is Left As An Exercise To The Reader. 😊

we show this unique “ γ ” with “ $\beta -_c \alpha$ ”

definition 10.4.1. Trunk and Branch

1. $\text{trunk}(\alpha) := \sup\{\beta \in \alpha \mid \beta +_c \alpha = \alpha\}$
2. $\text{branch}(\alpha) := \alpha -_c \text{trunk}(\alpha)$
3. $\text{trunk}(\alpha) := \begin{cases} \text{trunk}(\alpha) & \text{branch}(\alpha) < \beta \\ \text{trunk}(\alpha) +_c \text{trunk}(\text{branch}(\alpha, \beta)) & \text{branch}(\alpha) \geq \beta \end{cases}$
4. $\text{branch}(\alpha, \beta) := \alpha -_c \text{trunk}(\alpha, \beta)$

10.5 Arithmetic of Ordinals

Theorem 10.5.1. Summation *there is a unique function “ $+$: $\text{Ord} \times \text{Ord} \rightarrow \text{Ord}$ ” s.t.*

1. $\forall \alpha \in \text{Ord} \quad \alpha + 0 = \alpha$
2. $\forall \alpha, \beta \in \text{Ord} \quad \alpha + s(\beta) = s(\alpha + \beta)$
3. $\forall \alpha, \beta \in \text{Ord} \quad \text{limit}(\beta) \wedge \alpha < \beta \rightarrow \alpha + \beta = \beta + \alpha$
4. $\forall \alpha, \beta \in \text{Ord} \quad \text{limit}(\beta) \wedge \beta \leq \alpha \rightarrow \alpha + \beta = (\text{trunk}(\alpha, \beta) +_c \beta) + \text{branch}(\alpha, \beta)$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 10.5.2. Multiplication *there is a unique function “ $\times$: $\text{Ord} \times \text{Ord} \rightarrow \text{Ord}$ ” s.t.*

1. $\forall \alpha \in \text{Ord} \quad \alpha \times 0 = 0$
2. $\forall \alpha, \beta \in \text{Ord} \quad \alpha \times s(\beta) = \alpha \times \beta + \alpha$
3. $\forall \alpha, \beta \in \text{Ord} \quad \text{limit}(\beta) \wedge \alpha < \beta \rightarrow \alpha \times \beta = \beta \times \alpha$
4. $\forall \alpha, \beta \in \text{Ord} \quad \text{limit}(\beta) \wedge \beta \leq \alpha \rightarrow \alpha \times \beta = \bigcup_{\beta' \in \text{trunk}(\beta)} (\alpha \times \beta') + \alpha \times \text{branch}(\beta)$

Proof Is Left As An Exercise To The Reader. 😊

we denote also “Multiplication” with “ $\alpha \cdot \beta$ ” and “ $\alpha\beta$ ” too.

10.6 Hyper Peano arithmetic

assume a structure “ $\mathfrak{N} := (N, +, \cdot, s, 0)$ ”(here “ $+$, “ $\cdot$ ”, “ s ”, “ 0 ” are arbitrary functions and constants and doesn’t have meaning we defined before). with “ $\text{Limit}(x) : \forall y \in N \quad s(y) \neq x$ ” and these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \rightarrow x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$

4. $\forall x \in N \quad x \cdot 0 = 0$
5. $\forall x, y \in N \quad x + s(y) = s(x + y)$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\forall x, y \in N \quad Limit(y) \rightarrow x + y = y + x$
8. $\forall x, y \in N \quad Limit(y) \rightarrow xy = yx$
9. $\forall x, y, z \in N \quad Limit(z) \rightarrow (x + y) + z = y + (x + z)$
10. $\forall x, y, z \in N \quad Limit(z) \rightarrow (xy)z = y(xz)$
11. $\forall x, y, z \in N \quad Limit(z) \rightarrow x(y + z) = xy + xz$
12. $\forall x, y, z \in N \quad Limit(z) \rightarrow x + z = y + z \rightarrow x = y$
13. $\forall x, y \in N \quad Limit(y) \rightarrow xy = 0 \rightarrow x = 0 \vee y = 0$
14. $\left(\forall x \in N \quad Limit(x) \rightarrow \phi(x) \right) \wedge \left(\forall y \in N \quad (\phi(y) \rightarrow \phi(s(y))) \right) \rightarrow \forall y \in N \quad \phi(y)$ (this is true for arbitrary “ ϕ ”)

we call “ $\mathfrak{N}$ ” a model of “Hyper Peano arithmetic”.

Lemma 10.6.1. *prove for “ $(N, +)$ ”*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ (*assosiative +*)
2. $\forall x \in N \quad x + 0 = 0 + x = x$ (*identity +*)
3. $\forall x, y, z \in N \quad x + z = y + z \rightarrow x = y$ (*cancellative +*)
4. $\forall x, y \in N \quad x + y = y + x$ (*commutative(abelian) +*)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 10.6.2. *prove for “ $(N, +, \cdot)$ ”*

1. $\forall x, y, z \in N \quad (mn)z = x(nk)$ (*assosiative $\times$*)
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ (*identity $\times$*)
3. $\forall x, y \in N \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (*zero divisor free $\times$*)
4. $\forall x, y \in N \quad xy = yx$ (*commutative(abelian) $\times$*)
5. $\forall x, y, z \in N \quad \left(x(y + z) = xy + xz \right) \wedge \left((x + y)z = xz + yz \right)$ (*distributive $\times$*)

Proof Is Left As An Exercise To The Reader. 😊

Now define “ $x < y : \exists z \neq 0 \quad x + z = y$ ”. You can prove for natural numbers

$$\forall \alpha, \beta \in Ord \quad \alpha < \beta \leftrightarrow \exists k \neq 0 \quad \alpha + k = \beta$$

Lemma 10.6.3. *prove for “ $(N, +, \times, <)$ ”*

1. $\forall x, y, z \in N \quad x < y \rightarrow x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

as you can see “ $(Ord, +, \times, s, 0)$ ” is a model of hyper peano arithmetic. so properties are true for them.

This page intentionally left blank.

Chapter 11

Cardinals

This page intentionally left blank.

Chapter 12

Axiom of choice

This page intentionally left blank.

Chapter 13

hyperreal numbers $({}^*\mathbb{N}, {}^*\mathbb{Z}, {}^*\mathbb{Q}, {}^*\mathbb{R})$ and surreal numbers

1. Starting Point: Sequences of Reals

Consider the set of all sequences of real numbers:

$$\mathbb{R}^{\mathbb{N}} = \{(x_1, x_2, x_3, \dots) : x_n \in \mathbb{R}\}.$$

Think of a sequence as a function from $\mathbb{N}$ to $\mathbb{R}$.

2. Nonprincipal Ultrafilter on $\mathbb{N}$

An *ultrafilter* $\mathcal{U}$ on $\mathbb{N}$ is a collection of subsets of $\mathbb{N}$ satisfying:

1. If $A \in \mathcal{U}$ and $A \subseteq B \subseteq \mathbb{N}$, then $B \in \mathcal{U}$.
2. If $A, B \in \mathcal{U}$, then $A \cap B \in \mathcal{U}$.
3. For every $A \subseteq \mathbb{N}$, either $A \in \mathcal{U}$ or $\mathbb{N} \setminus A \in \mathcal{U}$.

An ultrafilter is *nonprincipal* if it contains no finite sets. Existence of such ultrafilters follows from the Axiom of Choice.

Intuitively, a nonprincipal ultrafilter determines what it means for something to happen “for almost all” n .

3. Equivalence Relation on Sequences

For two sequences (x_n) and (y_n) in $\mathbb{R}^{\mathbb{N}}$, define

$$(x_n) \sim (y_n) \iff \{n \in \mathbb{N} : x_n = y_n\} \in \mathcal{U}.$$

That is, they are considered equivalent if they agree on a “large” set of indices belonging to the ultrafilter.

4. The Ultrapower Construction

The set of equivalence classes under this relation is

$${}^*\mathbb{R} = \mathbb{R}^{\mathbb{N}} / \sim.$$

If (x_n) is a sequence, its equivalence class is denoted $[x_n]$.

5. Algebraic Operations

Addition and multiplication are defined coordinatewise:

$$[x_n] + [y_n] = [x_n + y_n], \quad [x_n] \cdot [y_n] = [x_n y_n].$$

These operations are well-defined on equivalence classes, and with them, ${}^*\mathbb{R}$ becomes a field.

6. Embedding of $\mathbb{R}$

Each real number $r \in \mathbb{R}$ defines a constant sequence $(r, r, r, \dots)$. Define the embedding

$$\iota : \mathbb{R} \rightarrow {}^*\mathbb{R}, \quad r \mapsto [r, r, r, \dots].$$

This is an injective field homomorphism, so we can view $\mathbb{R} \subset {}^*\mathbb{R}$.

7. Infinitesimal and Infinite Elements

Infinitesimal: Choose $x_n = \frac{1}{n}$. Then for every standard $k \in \mathbb{N}$,

$$\{n : \frac{1}{n} < \frac{1}{k}\}$$

is cofinite, hence belongs to $\mathcal{U}$. Therefore

$$0 < [1/n] < \frac{1}{k} \quad \forall k \in \mathbb{N}.$$

Thus $[1/n]$ is a positive infinitesimal.

Infinite element: Choose $y_n = n$. For any fixed $M \in \mathbb{N}$,

$$\{n : n > M\}$$

is cofinite, so $[n] > M$ in ${}^*\mathbb{R}$. Hence $[n]$ is an infinite hyperreal number.

8. The Transfer Principle (Łoś's Theorem)

Łoś's theorem: For any first-order formula $\varphi(x_1, \dots, x_k)$,

$${}^*\mathbb{R} \models \varphi([x_n^{(1)}], \dots, [x_n^{(k)}]) \quad \text{iff} \quad \{n \in \mathbb{N} : \mathbb{R} \models \varphi(x_n^{(1)}, \dots, x_n^{(k)})\} \in \mathcal{U}.$$

This yields the *Transfer Principle*: any first-order property true in $\mathbb{R}$ is also true in ${}^*\mathbb{R}$.

9. Summary

1. Start with all real sequences $\mathbb{R}^{\mathbb{N}}$.
2. Pick a nonprincipal ultrafilter $\mathcal{U}$ on $\mathbb{N}$.
3. Define $(x_n) \sim (y_n)$ if they agree on a set in $\mathcal{U}$.
4. The quotient $\mathbb{R}^{\mathbb{N}} / \sim$ is the hyperreal field ${}^*\mathbb{R}$.
5. Embed $\mathbb{R}$ via constant sequences.
6. Sequences like $1/n$ and n give infinitesimal and infinite elements.
7. Łoś's theorem provides the logical foundation (Transfer Principle).

10. Remarks

${}^*\mathbb{R}$ depends on the choice of the nonprincipal ultrafilter, but all such constructions are *elementarily equivalent* fields—they satisfy the same first-order properties of $\mathbb{R}$.

Common Symbols: ω , ϵ , ε , and Others

1. The Symbol ω

$$\omega = \{0, 1, 2, 3, \dots\}$$

$$|\omega| = \aleph_0$$

Ultrafilter and nonstandard analysis context:

$$[n] > k \quad \text{for all standard } k \in \mathbb{N}$$

LaTeX:

`\omega`

2. The Symbols ϵ and ε

$$\epsilon > 0, \quad \varepsilon > 0$$

Both denote an arbitrarily small positive real number.

Limit definition example:

$$\forall \varepsilon > 0 \exists \delta > 0 : |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Infinitesimal in hyperreals:

$$0 < \varepsilon < \frac{1}{n} \quad \forall n \in \mathbb{N}.$$

LaTeX:

`\epsilon`, `\varepsilon`

3. Other Common Greek Symbols

$$\alpha, \beta, \gamma, \delta, \eta, \theta, \lambda, \mu, \phi, \varphi$$

LaTeX:

`\alpha, \beta, \gamma, \delta, \eta, \theta, \lambda, \mu, \phi, \varphi.`

4. Summary Table (LaTeX Only)

Symbol	Meaning	LaTeX
ω	first infinite ordinal	<code>\omega</code>
Ω	asymptotic lower bound	<code>\Omega</code>
ϵ	small positive number	<code>\epsilon</code>
ε	alternative epsilon	<code>\varepsilon</code>
δ	small positive number in limits	<code>\delta</code>
θ	angle or parameter	<code>\theta</code>
λ	eigenvalue or parameter	<code>\lambda</code>

Chapter 14

fantasy

numbers($\mathbb{Z}(Ord)$, $\mathbb{Q}(Ord)$, $\mathbb{R}(Ord)$, $\mathbb{C}(Ord)$)

14.1 Hyper Integers“ $\mathbb{Z}(Ord)$ ”

we learned about “ Ord ” before and we constructed structure “ $(Ord, +, \cdot, <)$ ”. if you assume “ $Ord \times Ord$ ” we can define an equivalence relation “ $(\alpha_1, \beta_1) \sim (\alpha_2, \beta_2) : \alpha_1 + \beta_2 = \alpha_2 + \beta_1$ ”(check it’s an equivalence relation).

definition 14.1.1. *Hyper Integers“ $\mathbb{Z}(Ord)$ ” is a set*

$$\mathbb{Z}(Ord) := Ord \times Ord / \sim$$

we denote it’s members with “ $z_0, z_1, \dots$ ”

in this system “ $0 := [(0, 0)]$ ” and “ $\forall \beta \in Ord \quad \beta := [(\beta, 0)]$ ” and “ $\forall \beta \in Ord \quad -\beta := [(0, \beta)]$ ”. in general “ $\forall \alpha, \beta \in Ord \quad \alpha - \beta := [(\alpha, \beta)]$ ”. so we can say “ $Ord \subseteq \mathbb{Z}(Ord)$ ”.

Theorem 14.1.1. Summation *there is a unique function “ $+_{\mathbb{Z}(Ord)} : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \rightarrow \mathbb{Z}(Ord)$ ” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] +_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1 + \alpha_2, \beta_1 + \beta_2)]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.1.2. Multiplication *there is a unique function “ $\times_{\mathbb{Z}(Ord)} : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \rightarrow \mathbb{Z}(Ord)$ ” s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] \times_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1 \alpha_2 + \beta_1 \beta_2, \alpha_1 \beta_2 + \alpha_2 \beta_1)]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 14.1.2. Order in $\mathbb{Z}(Ord)$

$$[(\alpha_1, \beta_1)] <_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] : \alpha_1 + \beta_2 < \alpha_2 + \beta_1$$

this lemma let us use “ $<$ ” instead of “ $<_{\mathbb{Z}(\text{Ord})}$ ”, “ $+$ ” instead of “ $+_{\mathbb{Z}(\text{Ord})}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{Z}(\text{Ord})}$ ”.

Lemma 14.1.1. *prove.*

1. $\therefore \forall \alpha, \beta \in \text{Ord} \quad \alpha <_{\mathbb{Z}(\text{Ord})} \beta \leftrightarrow \alpha < \beta$
2. $\therefore \forall \alpha, \beta \in \text{Ord} \quad \alpha +_{\mathbb{Z}(\text{Ord})} \beta = \alpha + \beta$
3. $\therefore \forall \alpha, \beta \in \text{Ord} \quad \alpha \times_{\mathbb{Z}(\text{Ord})} \beta = \alpha \times \beta$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $z \cdot y$ ” and “ zy ” too.

Lemma 14.1.2. *prove for “ $(\mathbb{Z}(\text{Ord}), +)$ ”*

1. $\forall x, y, z \in \mathbb{Z}(\text{Ord}) \quad (x + y) + z = x + (y + z)$ (*assosiative* +)
2. $\forall z \in \mathbb{Z}(\text{Ord}) \quad z + 0 = 0 + z = z$ (*identity* +)
3. $\forall z \in \mathbb{Z}(\text{Ord}) \quad \exists y \in \mathbb{Z}(\text{Ord}) \quad z + y = y + z = 0$ (*inverse* +)
4. $\forall x, y \in \mathbb{Z}(\text{Ord}) \quad x + y = y + x$ (*commutative(abelian)* +)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.1.3. *prove for “ $(\mathbb{Z}(\text{Ord}), +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{Z}(\text{Ord}) \quad (xy)z = x(yz)$ (*assosiative* $\times$)
2. $\forall z \in \mathbb{Z}(\text{Ord}) \quad z \cdot 1 = 1 \cdot z = z$ (*identity* $\times$)
3. $\forall x, y \in \mathbb{Z}(\text{Ord}) \quad xy = 0 \rightarrow x = 0 \vee y = 0$ (*zero divisor free* $\times$)
4. $\forall x, y \in \mathbb{Z}(\text{Ord}) \quad xy = yx$ (*commutative(abelian)* $\times$)
5. $\forall x, y, z \in \mathbb{Z}(\text{Ord}) \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (*distributive* $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.1.3. Order of Hyper Integers($\mathbb{Z}(\text{Ord})$) *prove for “ $(\mathbb{Z}(\text{Ord}), +, \times, <_{\mathbb{Z}(\text{Ord})})$ ”*

1. *is lineary ordered.*
2. $(\forall x \in \mathbb{Z}(\text{Ord}) \quad \exists y \in \mathbb{Z}(\text{Ord}) \quad x < y) \wedge (\forall x \in \mathbb{Z}(\text{Ord}) \quad \exists y \in \mathbb{Z}(\text{Ord}) \quad y < x)$ (*doesn't have endpoints*)
3. $\forall x, y, z \in \mathbb{Z}(\text{Ord}) \quad x < y \rightarrow x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z}(\text{Ord}) \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

14.2 Hyper Rational numbers “ $\mathbb{Q}(\text{Ord})$ ”

we learned about “ $\mathbb{Z}(\text{Ord})$ ” before and we constructed structure “ $(\mathbb{Z}(\text{Ord}), +, \cdot, <)$ ”. if you assume “ $\mathbb{Z}(\text{Ord}) \times (\mathbb{Z}(\text{Ord}) - \{0\})$ ” we can define an equivalence relation “ $(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$ ”(check it's an equivalence relation).

definition 14.2.1. Hyper Rational numbers “ $\mathbb{Q}(\text{Ord})$ ” is a set

$$\mathbb{Q}(\text{Ord}) := \mathbb{Z}(\text{Ord}) \times (\mathbb{Z}(\text{Ord}) - \{0\}) / \sim$$

we denote it's members with “ $q_0, q_1, \dots$ ”

in this system “ $0 := [(0, 1)]$ ” and “ $1 := [(1, 1)]$ ” and “ $\forall x \in \mathbb{Z}(\text{Ord}) \quad x := [(x, 1)]$ ” and “ $\forall x \in \mathbb{Z}(\text{Ord}) \quad \frac{1}{x} := [(1, x)]$ ”. in general “ $\forall x, y \in \mathbb{Z}(\text{Ord}) \quad \frac{x}{y} := [(x, y)]$ ”. so we can say “ $\text{Ord} \subseteq \mathbb{Z}(\text{Ord}) \subseteq \mathbb{Q}(\text{Ord})$ ”.

Theorem 14.2.1. Summation there is a unique function “ $+_{\mathbb{Q}(\text{Ord})} : \mathbb{Q}(\text{Ord}) \times \mathbb{Q}(\text{Ord}) \rightarrow \mathbb{Q}(\text{Ord})$ ” s.t.

$$\forall x, y, z, u \in \mathbb{Z}(\text{Ord}) \quad [(x, z)] +_{\mathbb{Q}(\text{Ord})} [(y, u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.2.2. Multiplication there is a unique function “ $\times_{\mathbb{Q}(\text{Ord})} : \mathbb{Q}(\text{Ord}) \times \mathbb{Q}(\text{Ord}) \rightarrow \mathbb{Q}(\text{Ord})$ ” s.t.

$$\forall x, y, z, u \in \mathbb{Z}(\text{Ord}) \quad [(x, z)] \times_{\mathbb{Q}(\text{Ord})} [(y, u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 14.2.2. Order in $\mathbb{Q}(\text{Ord})$.

if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x, z)] <_{\mathbb{Q}(\text{Ord})} [(y, u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x, z)] <_{\mathbb{Q}(\text{Ord})} [(y, u)] : x \cdot u > y \cdot z$$

this lemma let us use “ $<$ ” instead of “ $<_{\mathbb{Q}(\text{Ord})}$ ”, “ $+$ ” instead of “ $+_{\mathbb{Q}(\text{Ord})}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{Q}(\text{Ord})}$ ”.

Lemma 14.2.1. prove.

1. / $\therefore \forall z, y \in \mathbb{Z}(\text{Ord}) \quad z <_{\mathbb{Q}(\text{Ord})} y \leftrightarrow z < y$
2. / $\therefore \forall z, y \in \mathbb{Z}(\text{Ord}) \quad z +_{\mathbb{Q}(\text{Ord})} y = z + y$
3. / $\therefore \forall z, y \in \mathbb{Z}(\text{Ord}) \quad z \times_{\mathbb{Q}(\text{Ord})} y = z \times y$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $x \cdot y$ ” and “ xy ” too.

Lemma 14.2.2. prove for “ $(\mathbb{Q}(\text{Ord}), +)$ ”

1. $\forall x, y, z \in \mathbb{Q}(\text{Ord}) \quad (x + y) + z = x + (y + z)$ (assosiative +)
2. $\forall x \in \mathbb{Q}(\text{Ord}) \quad x + 0 = 0 + x = x$ (identity +)
3. $\forall x \in \mathbb{Q}(\text{Ord}) \quad \exists y \in \mathbb{Q}(\text{Ord}) \quad x + y = y + x = 0$ (inverse +)

4. $\forall x, y \in \mathbb{Q}(\text{Ord}) \quad x + y = y + x$ (commutative(abelian) $+$)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.2.3. *prove for “ $(\mathbb{Q}(\text{Ord}), +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{Q}(\text{Ord}) \quad (xy)z = x(yz)$ (assosiative $\times$)
2. $\forall x \in \mathbb{Q}(\text{Ord}) \quad x \cdot 1 = 1 \cdot x = x$ (identity $\times$)
3. $\forall x \in \mathbb{Q}(\text{Ord}) - \{0\} \quad \exists y \in \mathbb{Q}(\text{Ord}) \quad xy = yx = 1$ (inverse $\times$)
4. $\forall x, y \in \mathbb{Q}(\text{Ord}) \quad xy = yx$ (commutative(abelian) $\times$)
5. $\forall x, y, z \in \mathbb{Q}(\text{Ord}) \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.2.3. Order of Hyper Rational numbers($\mathbb{Q}(\text{Ord})$) *prove for “ $(\mathbb{Q}(\text{Ord}), <_{\mathbb{Q}(\text{Ord})})$ ”*

1. *is lineary ordered.*
2. $(\forall x \in \mathbb{Q}(\text{Ord}) \quad \exists y \in \mathbb{Q}(\text{Ord}) \quad x < y) \wedge (\forall x \in \mathbb{Q}(\text{Ord}) \quad \exists y \in \mathbb{Q}(\text{Ord}) \quad y < x)$ (doesn't have endpoints)
3. $\forall x, y \in \mathbb{Q}(\text{Ord}) \quad \exists z \in \mathbb{Q}(\text{Ord}) \quad x \neq y \rightarrow x < z < y$ (dense)
4. $\forall x, y, z \in \mathbb{Q}(\text{Ord}) \quad x < y \rightarrow x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q}(\text{Ord}) \quad x < y \wedge 0 < z \rightarrow xz < yz$

Proof Is Left As An Exercise To The Reader. 😊

14.3 Hyper Real numbers “ $\mathbb{R}(\text{Ord})$ ”

before we we get to lesson define function “ $|_| : \mathbb{Q}(\text{Ord}) \times \mathbb{Q}(\text{Ord}) \rightarrow \mathbb{Q}(\text{Ord})$ ” like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means “ $|_| := \{(x, y) \in \mathbb{Q}(\text{Ord}) \times \mathbb{Q}(\text{Ord}) \mid (0 \leq x \rightarrow y = x) \wedge (x < 0 \rightarrow y = -x)\}$ ”. learn this notation because we want to use later.

we learned about “ $\mathbb{Q}(\text{Ord})$ ” before and we constructed structure “ $(\mathbb{Q}(\text{Ord}), +, \cdot, <)$ ”. now assume $\mathbb{Q}(\text{Ord})^\omega$ (we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}(\text{Ord})$). we don't want it all we just want a subset.

definition 14.3.1. Cauchy sequence *is a sequence like “ $(a_i)_{i \in \omega} \in \mathbb{Q}(\text{Ord})^\omega$ ” s.t. elements of it get infinitely close to each other i.e.*

$$\text{Cauchy}(a_i)_{i \in \omega} : \forall \epsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \epsilon$$

we usually give “ N ” natural numbers but it’s not necessary.

we define an equivalence relation “ $(a_i)_{i \in \omega} \sim (b_i)_{i \in \omega} : \forall \epsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \epsilon$ ” (check it’s an equivalence relation).

definition 14.3.2. Hyper Real numbers “ $\mathbb{R}(Ord)$ ” is a set

$$\mathbb{R}(Ord) := \text{Cauchy} / \sim$$

we denote it’s members with “ $r_0, r_1, \dots$ ”

in this system “ $0 := [(0)_{i \in \omega}]$ ” and “ $1 := [(1)_{i \in \omega}]$ ” and “ $\forall q \in \mathbb{Q}(Ord) \quad q := [(q)_{i \in \omega}]$ ”. so we can say “ $Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord)$ ”.

Theorem 14.3.1. Summation there is a unique function “ $+_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \rightarrow \mathbb{R}(Ord)$ ” s.t.

$$[(a_i)_{i \in Ord}] +_{\mathbb{R}(Ord)} [(b_i)_{i \in Ord}] = [(a_i + b_i)_{i \in Ord}]$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.3.2. Multiplication there is a unique function “ $\times_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \rightarrow \mathbb{R}(Ord)$ ” s.t.

$$[(a_i)_{i \in Ord}] \times_{\mathbb{R}(Ord)} [(b_i)_{i \in Ord}] = [(a_i b_i)_{i \in Ord}]$$

Proof Is Left As An Exercise To The Reader. 😊

definition 14.3.3. Order in $\mathbb{R}(Ord)$.

$$[(a_i)_{i \in Ord}] <_{\mathbb{R}(Ord)} [(b_i)_{i \in Ord}] : \exists \epsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \epsilon$$

this lemma let us use “ $<$ ” instead of “ $<_{\mathbb{R}(Ord)}$ ”, “ $+$ ” instead of “ $+_{\mathbb{R}(Ord)}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{R}(Ord)}$ ”.

Lemma 14.3.1. *prove.*

1. $\therefore \forall x, y \in \mathbb{Q}(Ord) \quad x <_{\mathbb{R}(Ord)} y \leftrightarrow x < y$
2. $\therefore \forall x, y \in \mathbb{Q}(Ord) \quad x +_{\mathbb{R}(Ord)} y = x + y$
3. $\therefore \forall x, y \in \mathbb{Q}(Ord) \quad x \times_{\mathbb{R}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $x \cdot y$ ” and “ xy ” too.

Lemma 14.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (x + y) + z = x + (y + z)$ (*assosiative +*)
2. $\forall x \in \mathbb{R}(Ord) \quad x + 0 = 0 + x = x$ (*identity +*)
3. $\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x + y = y + x = 0$ (*inverse +*)
4. $\forall x, y \in \mathbb{R}(Ord) \quad x + y = y + x$ (*commutative(abelian) +*)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R}(\text{Ord}) \quad (xy)z = x(yz)$ (assosiative $\times$)
2. $\forall x \in \mathbb{R}(\text{Ord}) \quad x \cdot 1 = 1 \cdot x = x$ (identity $\times$)
3. $\forall x \in \mathbb{R}(\text{Ord}) - \{0\} \quad \exists y \in \mathbb{R}(\text{Ord}) \quad xy = yx = 1$ (inverse $\times$)
4. $\forall x, y \in \mathbb{R}(\text{Ord}) \quad xy = yx$ (commutative(abelian) $\times$)
5. $\forall x, y, z \in \mathbb{R}(\text{Ord}) \quad \left(x(y+z) = xy + xz \right) \wedge \left((x+y)z = xz + yz \right)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.3.3. Linear order of Hyper Real numbers($\mathbb{R}(\text{Ord})$) *prove “($\mathbb{R}(\text{Ord})$, $<_{\mathbb{R}(\text{Ord})}$)*

1. *is lineary ordered.*
2. $\left(\forall x \in \mathbb{R}(\text{Ord}) \quad \exists y \in \mathbb{R}(\text{Ord}) \quad x < y \right) \wedge \left(\forall x \in \mathbb{R}(\text{Ord}) \quad \exists y \in \mathbb{R}(\text{Ord}) \quad y < x \right)$ (doesn't have endpoints)
3. $\forall x, y \in \mathbb{R}(\text{Ord}) \quad \exists z \in \mathbb{R}(\text{Ord}) \quad x \neq y \rightarrow x < z < y$ (dense)
4. $\forall X \subseteq \mathbb{R}(\text{Ord}) \quad (X \neq \emptyset \wedge \exists y \in \mathbb{R}(\text{Ord}) \quad \text{Uppreb}(X, y)) \rightarrow (\exists y \in \mathbb{R}(\text{Ord}) \quad \text{sup}(X) = y)$ (complete)
5. $\forall x, y, z \in \mathbb{R}(\text{Ord}) \quad x < y \rightarrow x + z < y + z$
6. $\forall x, y, z \in \mathbb{R}(\text{Ord}) \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader. 😊

14.4 Hyper Complex numbers“ $\mathbb{C}(\text{Ord})$ ”

we learned about “ $\mathbb{R}(\text{Ord})$ ” before and we constructed structure “($\mathbb{R}(\text{Ord})$, $+$, $\cdot$, $<$)”.

definition 14.4.1. Hyper Complex numbers“ $\mathbb{C}(\text{Ord})$ ” *is a set*

$$\mathbb{C}(\text{Ord}) := \mathbb{R}(\text{Ord}) \times \mathbb{R}(\text{Ord})$$

we denote it's members with “ $z_0, x, \dots$ ”

in this system “ $0 := (0, 0)$ ” and “ $\forall n \in \mathbb{R}(\text{Ord}) \quad r := (r, 0)$ ” . in general “ $\forall x, y \in \mathbb{R}(\text{Ord}) \quad x + iy := (x, y)$ ”. so we can say “ $\text{Ord} \subseteq \mathbb{Z}(\text{Ord}) \subseteq \mathbb{Q}(\text{Ord}) \subseteq \mathbb{R}(\text{Ord}) \subseteq \mathbb{C}(\text{Ord})$ ”.

Theorem 14.4.1. Summation *there is a unique function “ $+_{\mathbb{C}(\text{Ord})} : \mathbb{C}(\text{Ord}) \times \mathbb{C}(\text{Ord}) \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.*

$$\forall x, y, z, u \in \mathbb{R}(\text{Ord}) \quad (x, z) +_{\mathbb{C}(\text{Ord})} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.4.2. Multiplication there is a unique function “ $\times_{\mathbb{C}(\text{Ord})} : \mathbb{C}(\text{Ord}) \times \mathbb{C}(\text{Ord}) \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.

$$\forall x, y, z, u \in \mathbb{C}(\text{Ord}) \quad (x, z) \times_{\mathbb{C}(\text{Ord})} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader. 😊

this lemma let us use “+” instead of “ $+_{\mathbb{C}(\text{Ord})}$ ” and “ $\times$ ” instead of “ $\times_{\mathbb{C}(\text{Ord})}$ ”.

Lemma 14.4.1. *prove.*

1. $\therefore \forall x, y \in \mathbb{C}(\text{Ord}) \quad x +_{\mathbb{C}(\text{Ord})} y = x + y$
2. $\therefore \forall x, y \in \mathbb{C}(\text{Ord}) \quad x \times_{\mathbb{C}(\text{Ord})} y = x \times y$

Proof Is Left As An Exercise To The Reader. 😊

we denote “Multiplication” with “ $z \cdot y$ ” and “ zy ” too.

Lemma 14.4.2. *prove for “ $(\mathbb{C}(\text{Ord}), +)$ ”*

1. $\forall x, y, z \in \mathbb{C}(\text{Ord}) \quad (x + y) + z = x + (y + z)$ (assosiative +)
2. $\forall z \in \mathbb{C}(\text{Ord}) \quad z + 0 = 0 + z = z$ (identity +)
3. $\forall z \in \mathbb{C}(\text{Ord}) \quad \exists y \in \mathbb{C}(\text{Ord}) \quad z + y = y + z = 0$ (inverse +)
4. $\forall x, y \in \mathbb{C}(\text{Ord}) \quad x + y = y + x$ (commutative(abelian) +)

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.4.3. *prove for “ $(\mathbb{C}(\text{Ord}), +, \times)$ ”*

1. $\forall x, y, z \in \mathbb{C}(\text{Ord}) \quad (xy)z = x(yz)$ (assosiative $\times$)
2. $\forall z \in \mathbb{C}(\text{Ord}) \quad z \cdot 1 = 1 \cdot z = z$ (identity $\times$)
3. $\forall x, y \in \mathbb{C}(\text{Ord}) \quad xy = yx$ (commutative(abelian) $\times$)
4. $\forall x, y, z \in \mathbb{C}(\text{Ord}) \quad (x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)$ (distributive $\times$)

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.4.3. Exponents there is a unique function “ $(_)^{(_)} : \mathbb{C}(\text{Ord}) \times \text{Ord} \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.

1. $\forall x \in \mathbb{C}(\text{Ord}) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall n \in \text{Ord} \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.4.4. *prove for “ $(\mathbb{C}(\text{Ord}), +, \cdot, (_)^{(_)})$ ”*

1. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall m, n \in \text{Ord} \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall m, n \in \text{Ord} \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.4.4. Exponents there is a unique function “ $(_)^{(_)} : \mathbb{C}(\text{Ord}) \times \mathbb{Z}(\text{Ord}) \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.

1. $\forall x \in \mathbb{C}(\text{Ord}) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall m, n \in \text{Ord} \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.4.5. prove for “ $(\mathbb{C}(\text{Ord}), +, \cdot)$ ”

1. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall y, z \in \mathbb{Z}(\text{Ord}) \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C}(\text{Ord}) \quad \forall y, z \in \mathbb{Z}(\text{Ord}) \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.4.5. Big Sum if “ $a : \text{Ord} \rightarrow \mathbb{C}(\text{Ord})$ ” there is a unique function “ $\sum_{i=0}^{(_)}(_) : \mathbb{C}(\text{Ord}) \times \text{Ord} \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.

1. $\forall a \quad \sum_{i=0}^0(a_0) = a_0$
2. $\forall a \quad \forall n \in \text{Ord} \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^n a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.4.6. prove for “ $(\mathbb{C}(\text{Ord}), +, \cdot)$ ”

1. $\forall n \in \text{Ord} \quad \forall a, b \quad \sum_{i=0}^n a_n + \sum_{i=0}^n b_n = \sum_{i=0}^n (a_n + b_n)$
2. $\forall n \in \text{Ord} \quad \forall a \quad k \sum_{i=0}^n a_n = \sum_{i=0}^n k a_n$

Proof Is Left As An Exercise To The Reader. 😊

Theorem 14.4.6. Big Product if “ $a : \text{Ord} \rightarrow \mathbb{C}(\text{Ord})$ ” there is a unique function “ $\prod_{i=0}^{(_)}(_) : \mathbb{C}(\text{Ord}) \times \text{Ord} \rightarrow \mathbb{C}(\text{Ord})$ ” s.t.

1. $\forall a \quad \prod_{i=0}^0(a_0) = a_0$
2. $\forall a \quad \forall n \in \text{Ord} \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^n a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader. 😊

Lemma 14.4.7. prove for “ $(\mathbb{C}(\text{Ord}), +, \cdot)$ ”

1. $\forall n \in \text{Ord} \quad \forall a, b \quad \prod_{i=0}^n a_n \prod_{i=0}^n b_n = \prod_{i=0}^n (a_n b_n)$
2. $\forall n \in \text{Ord} \quad \forall a \quad (\prod_{i=0}^n a_n)^k = \prod_{i=0}^n (a_n)^k$

Proof Is Left As An Exercise To The Reader. 😊

Chapter 15

New beginning for calculus

15.1 Derivatives

first we have to fix a “Main variable” it’s almost all the time “ x ”. (sontimes it’s “ t ”).

definition 15.1.1. *assume $f : \mathbb{R}(Ord) \rightarrow \mathbb{R}(Ord)$ we define **Difrential** ($d_\alpha(f)$)*

$$d_\alpha(f) := f(x + \alpha) - f(x)$$

Main reason we choose “ x ” in defintion becuase it’s “Main variable”. you can prove $\forall x \in \mathbb{R}(Ord) \quad d_\alpha(x) = \alpha(x)$.

definition 15.1.2. *we define **Derivative***

$$\frac{d}{dx} := \{(f, sh(\frac{d_\alpha(f)}{d_\alpha(x)})) \mid f : \mathbb{R}(Ord) \rightarrow \mathbb{R}(Ord)\}$$

there is a biggest set like A s.t. $\frac{d}{dx} A$ is a function (you can prove it as an exercise). we call that set “Derivatible funsions” and that function “Derivative”.

This page intentionally left blank.

Chapter 16

Lebesgue Integral

This page intentionally left blank.

Chapter 17

New beginning for Probability theory